Identifying Rational Types in Unknown Environments under an Indirect Dynamic Mechanism⋆

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This paper develops a dynamic indirect mechanism to identify rational types in unknown environments characterized by observational pooling. The central theoretical contribution introduces the Mano de Dios–Guadalupe (MDG) process, an endogenous implementation noise kernel that maps latent intentions into executed actions with full support in the Bayesian game. Integrating the MDG process with a competing risks framework, the Rational Type Identification Theorem is established, whereby posterior Bayesian beliefs converge almost surely to the true type. Furthermore, the Implementability Theorem is presented, which guarantees that the mechanism is executable in a perfect Bayesian equilibrium. The empirical application to kidnapping for ransom in Colombia validates the theoretical framework through a microeconometric calibration and structural simulations that demonstrate the robustness of both Bayesian learning and incentive compatibility constraint compliance.

Keywords: indirect mechanism design, dynamic screening, endogenous implementation noise, Bayesian learning, type identification

JEL classification: C73, D82, D86

1. Introduction and Related Literature

The theory of mechanism design examines how a principal, understood as a planner, sets rules to implement a collective decision or resource allocation in environments where agents hold private information about their types and act strate-

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gically. In the classical framework established by Gibbard (1973) and Myerson (1979, 1981), the Revelation Principle postulates that any incentive-compatible outcome derived from an indirect mechanism can be represented by an equivalent direct mechanism in which truth-telling is the optimal strategy. This equivalence, however, is in some cases difficult to achieve in practice. As Bergemann and Morris (2012) note, mechanisms that are optimal in theory are frequently too complex or too sensitive to the details of the environment to be usable in practice. Compounding this informational fragility are agents' behavioral limitations; as Zhang and Levin (2017) show, bounded rationality and agents' psychological inability to evaluate contingencies state by state produce systematic deviations even from theoretically dominant strategies.

Therefore, when agents cannot or are not willing to declare their type directly, whether because of the mechanism's complexity, noise in observed actions, or because truthful revelation is not credible, the economically relevant object remains the *indirect* mechanism, grounded in observable actions and implementable, robust instruments. This paper develops that mechanism for dynamic environments with pooling equilibria, in which distinct types generate observationally equivalent short-run signals. A myopic policy, focused solely on minimizing immediate loss, can then perpetuate that pooling and prevent the asymptotic identification of the true types.

The paper is situated at the frontier of dynamic mechanism design, expanding the foundational theoretical frameworks of Courty and Li (2000), Eső and Szentes (2007), and Pavan et al. (2014). As also pointed out and complemented by the survey of Bergemann and Välimäki (2019), this work incorporates active informational exploration and structural noise, both of which are indispensable elements for making mechanism design viable in adversarial institutional contexts. This paper proposes to move beyond the static implementability frontier delimited by Myerson and Satterthwaite (1983), Rochet (1987), and Rochet and Choné (1998). To that end, and complementing the existence theory for static optimal contracts under direct revelation of Kadan et al. (2017), the central contribution consists in formulating and characterizing an indirect dynamic mechanism subject to endogenous implementation noise. This modeling introduces an intrinsic stochastic friction between the agent's latent strategic intention and its publicly recorded action, which makes it possible to guarantee both the Bayesian identification of rational types and conditional implementability under the perfect Bayesian equilibrium (PBE) criterion. Specifically, the paper structures this contribution through four interconnected theoretical results.

Indirect mechanism. The paper formalizes a dynamic mechanism in which the principal chooses instruments as functions of the public history and beliefs, while private agents best-respond optimally within their information sets. Types are not reported; they are inferred from the law of executed actions and outcomes. Thus, the Revelation Principle is preserved as a representation theorem, but the operative design object, and the paper's institutional contribution, is the indirect rule itself.

MDG endogenous implementation kernel. The Mano de Dios–Guadalupe (MDG) process, which is equivalent to an endogenous implementation kernel, maps latent intentions a_t^{i*} into executed actions $\tilde{a}_t^i$ through a hybrid-temperature logit with full support. Unlike classical refinements that inject exogenous trembles into strategies, as proposed by Selten (1975) and Kreps and Wilson (1982), MDG treats implementation noise as an intrinsic friction of materialization. Under a positive residual floor, the kernel maintains absolute continuity in the sense of Kalai and

Lehrer (1993), keeping Bayesian updating well defined on every branch while concentrating mass on the intended action. A maximum-entropy characterization in the spirit of Jaynes (1957) disciplines the functional form.

Type Identification Theorem for rational types. When each type induces a structural fingerprint that separates daily observation laws uniformly in total variation, the product-measure criterion of Kakutani (1948) yields mutual singularity of trajectory laws. Combined with the martingale stability of beliefs, as shown by Doob (1953) and Blackwell and Dubins (1962), posterior mass concentrates almost surely on the true type on the prolonged-interaction event $\mathcal{T} = \{\tau = \infty\}$. The theorem converts a local daily separation condition into global identification. Its limitation is explicit, since asymptotic consistency is stated on $\mathcal{T}$; finite-horizon simulations in the application illustrate directional learning without substituting for the asymptotic result.

Conditional PBE implementability. Identification alone does not make a mechanism executable. The Conditional Mechanism Implementability Theorem shows that, whenever the feasible set defined by incentive compatibility and individual rationality remains non-empty, there exists a PBE in which the principal's optimal rule implements the social choice subject to those constraints. Thus MDG, learning, and incentive feasibility form a single architecture: noise enables off-path updating; separation enables identification; and non-empty $\Gamma_t(\mu_t)$ enables equilibrium implementation.

The remainder of the article proceeds as follows. Section 2 presents the general environment, the indirect mechanism, the MDG kernel, competing risks, and best responses for arbitrary unknown types, and includes the Type Identification Theorem for Rational Types. Section 3 establishes conditional PBE implementability and discusses policy implications. Section 4 specializes the model to kidnapping in Colombia and empirically anchors the parameter space. Section 5 calibrates the mechanism and reports the baseline simulations. Section 6 assesses the robustness of those results under the counterfactual exclusion of the voice channel, isolating the contribution of physical-institutional competing risks to type identification. Section 7 discusses the results and concludes. The Supplemental Appendices contain the microeconometric detail (A), the formal proofs (B), and the technical documentation of the simulation engine (C).

2. General Environment and Indirect Mechanism

This section develops the theoretical model in general form. A principal designs an indirect dynamic mechanism when an agent's private type is unknown and equilibrium actions may pool. Types need not be revealed by direct reports; the mechanism operates on public histories, executed actions, and instruments. The Revelation Principle of Gibbard (1973) and Myerson (1979, 1981) continues to imply that any outcome implementable by an indirect mechanism admits an incentive-compatible direct representation; the contribution here is to construct and analyze the indirect object itself when messages are restricted to materialized actions subject to endogenous implementation noise.

2.1. Definition of the mechanism

Let $\mathcal{J} = \{S, K, F\}$ be the set of players, where S is the principal, state; K is the privately informed agent, captor; and F is a complementary agent, family. Each episode is characterized by the vector of types $\theta = (\theta_K, \theta_F, \theta_V, \theta_S) \in \Theta$, where the strategically private component θ_K takes values in a finite menu. In the Colombian application of Section 4, $\Theta_K = \{\text{DC}, \text{PAR}, \text{ELN}, \text{FARC}\}$ indexes distinct criminal technologies, representing common delinquency, paramilitary groups, the National Liberation Army, and the Revolutionary Armed Forces of Colombia, but the theorems treat Θ_K as an arbitrary finite type space; $\Theta_F = \{\text{High}, \text{Low}\}$ represents the family's payment capacity; $\Theta_V = \{\text{Public}, \text{Private}\}$ represents the victim's profile; and $\Theta_S = \{\text{Hard}, \text{Lax}\}$ represents the institutional environment. Each episode is also localized in an observable geographic zone $z \in \mathcal{R}$, where $\mathcal{R}$ groups five regions, Metropolitan, Andean, Caribbean, Pacific, and East-Jungle; z is not a private strategic type but a public initial condition that disciplines the prior over the captor's type. Only θ_K is private information of K; the other components and z are observable upon opening the episode, with an initial belief $\mu_0(\theta \mid z, \theta_V) = \mathbb{P}(\theta_K = \theta \mid z, \theta_V) \in \Delta(\Theta_K)$.

The public history $h_t \in H_t$ recursively records, up to the beginning of t and without including the type vector θ or the belief μ_t, the following elements: t is the period index; $\alpha^*_{[0,t-1]}$ and $\gamma^*_{[0,t-1]}$ are the trajectories of the state's financial interdiction and operational pressure instruments; $\tilde{a}^F_{[0,t-1]}$, $\tilde{a}^K_{[0,t-1]}$, and $\tilde{a}^S_{[0,t-1]}$ are the actions executed by the family, the captor, and the state; $m_{[0,t-1]}$ is the sequence of physical outcomes; and $d_{[0,t-1]}$ is the sequence of institutional detection signals, where $d_t \in \{0, 1\}$ indicates whether in period t the state detected collusion between the family and the captor, i.e., a clandestine ransom payment bypassing the financial interdiction policy.

$$h_t = \big(t,\, \alpha^*_{[0,t-1]},\, \gamma^*_{[0,t-1]},\, \tilde{a}^F_{[0,t-1]},\, \tilde{a}^K_{[0,t-1]},\, \tilde{a}^S_{[0,t-1]},\, m_{[0,t-1]},\, d_{[0,t-1]}\big), \tag{1}$$

$$h_{t+1} = \big(h_t,\, \alpha^*_t,\, \gamma^*_t,\, \tilde{a}^F_t,\, \tilde{a}^K_t,\, \tilde{a}^S_t,\, m_t,\, d_t\big). \tag{2}$$

The information sets when choosing in t are

$$\mathcal{I}_t^S = \big(h_t, z, \theta_F, \theta_V, \theta_S, \mu_t\big),$$
$$\mathcal{I}_t^F = \big(h_t, z, \theta_F, \theta_V, \theta_S\big), \qquad \mathcal{I}_t^K = \big(h_t, z, \theta_K, \theta_F, \theta_V, \theta_S\big).$$

Although μ_t is internal to the state, its exclusion from the private information sets does not generate inconsistency, because in the PBE private agents share the same prior and the same public history h_t, and therefore replicate the Bayesian updating of the planner Fudenberg and Tirole (1991a).

The space of institutional allocations of the state is $\mathcal{X}_t = \mathcal{A}^{S,\text{disc}} \times \mathcal{A}^{\alpha} \times \mathcal{A}^{\gamma}$. In each period t, players formulate optimal strategic intentions a_t^{i*}; the MDG process transforms them into executed actions $\tilde{A}_t = (\tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S)$ that enter the public registry and determine h_{t+1} and the public signal ζ_t. Conditional on the executed actions and the optimal institutional policy, the system determines the physical outcome $m_t \in \mathcal{M}_{\text{out}}$ through a block of competing risks, with $\mathcal{M}_{\text{cont}} = \{\text{cont}\}$, where cont denotes the continuation of captivity, $\mathcal{M}_{\text{term}} = \{\text{pay}, \text{res}, \text{rel}, \text{kill}\}$, which groups the termination causes: ransom payment, tactical rescue, voluntary release, and death of the victim, respectively, and $\mathcal{M}_{\text{out}} = \mathcal{M}_{\text{cont}} \cup \mathcal{M}_{\text{term}}$.

The horizon of the episode is not fixed in advance but is determined by the captivity dynamics themselves, in which the episode concludes in the first period in which a terminal outcome materializes, so the effective horizon is the endogenous stopping time.

$$\tau = \inf\{t \geq 0 : m_t \in \mathcal{M}_{\text{term}}\}, \tag{3}$$

where the recursive construction of h_t ensures that the event $\{\tau = t\}$ is measurable with respect to the public history at the beginning of t, guaranteeing that the state recognizes the closure of the episode in real time. On these elements, with $\mathbb{T} =$

$\{0, 1, \ldots, \tau\}$ as the active periods index, $\Theta = \Theta_K \times \Theta_F \times \Theta_V \times \Theta_S$ the type space, $(H_t)_{t\in\mathbb{T}}$ the family of public history spaces recursively generated by (1)–(2), $(\mu_t)_{t\in\mathbb{T}}$ the belief process in $\Delta(\Theta_K)$ updated by Bayes' rule throughout the episode, and $\mathbb{P}$ the ex-ante law induced by the strategy profile s, the prior μ_0, and the captivity dynamics, the dynamic indirect mechanism is formally defined as the tuple

$$\mathcal{D} = \big\langle \mathcal{J},\ \mathbb{T},\ \Theta,\ \mathcal{R},\ (\mathcal{X}_t)_{t\in\mathbb{T}},\ (H_t)_{t\in\mathbb{T}},\ (\mu_t)_{t\in\mathbb{T}},\ \mathbb{P}\big\rangle.$$

Given $z \in \mathcal{R}$ and $\theta_K \sim \mu_0$, the principal's strategy $s^S = \chi = (\chi_t)_{t\in\mathbb{T}}$ is implemented via the measurable public policy rule $\chi_t \colon H_t \times \mathcal{R} \to \mathcal{A}^{S,\text{disc}} \times \mathcal{A}^\alpha \times \mathcal{A}^\gamma$, which interacts with the private responses of K and F in their information sets; the social choice rule is formalized by the measurable mapping $f_t \colon \mathcal{I}_t^S \to \mathcal{X}_t$, such that the policy vector in equilibrium is $x_t^* := f_t(\mathcal{I}_t^S) = (a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \mathcal{X}_t$. This rule is implementable if it constitutes the path of a PBE in the sense of Fudenberg and Tirole (1991a), which guarantees that the informed agent cannot systematically induce the principal to calibrate instruments on a type that is not the true one; implementability is achieved when the pair (s, μ), with $s = (s^S, s^K, s^F)$ and $\mu_t \in \Delta(\Theta_K)$, satisfies two conditions in every reachable information set: *sequential rationality*, whereby each agent $i \in \{K, F\}$ maximises $\mathcal{U}_t^i$ given $\mathcal{I}_t^i$ and the strategies of the others while the state minimizes the expected social loss weighted by μ_t; and *Bayesian consistency*, whereby, in every node reached with positive probability under the measure $\mathbb{P}^s$ induced by the strategy profile s, the beliefs satisfy

$$\mu_t(\theta) = \mathbb{P}^s\big(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}}\big), \qquad \mathcal{F}_t^{\text{pub}} = \sigma\big(z, \zeta_0, \ldots, \zeta_{t-1}\big),$$

where $\mathcal{F}_t^{\text{pub}}$ is the public filtration at the beginning of t, generated by the σ-algebra of the initial geographic location z and the public signals $\zeta_0, \ldots, \zeta_{t-1}$; each signal $\zeta_u = (\alpha_u^*, \gamma_u^*, \tilde{a}_u^F, \tilde{a}_u^K, \tilde{a}_u^S, m_u, d_u)$ aggregates all observables of period u and constitutes the informational support over which the state updates its beliefs about θ_K; so that the instruments (α_t^*, γ_t^*) are calibrated on correct inferences about the captor's type.

2.2. *Summary of the mechanism*

In each active period t, the protocol advances in three sequential stages. In the first, the agents formulate their optimal intentions, as the state resolves $x_t^* = f_t(\mathcal{I}_t^S)$ and private agents select their best responses in their respective information sets, conditioned on the observable public policy (α_t^*, γ_t^*) signaled by the state. In the second, the MDG process transforms these intentions into executed actions $\tilde{A}_t$, guaranteeing full support across all branches of the game. In the third, the material state of the episode

$$\mathcal{C}_t(\theta_K) = \big(\theta_K, \theta_F, \theta_V, \theta_S, z, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, \alpha_t^*, \gamma_t^*, p_{\mathrm{det},t}\big),$$

with $p_{\mathrm{det},t}$ as the logit probability of detecting collusion, which is increasing in the state's instruments. This material state feeds a block of competing risks that determines m_t and d_t; the public signal ζ_t aggregates all observables of the period and updates the history to h_{t+1} and the belief to μ_{t+1} through Bayes' rule.

2.3. *Probabilistic technology*

This section formalizes the stochastic architecture that transforms strategic plans into observable realizations. In the base specification, the MDG process takes as input the latent strategic intention a_t^{i*} and generates an executed action $\tilde{a}_t^i$ through an implementation law concentrated on the plan, but with full support over $\mathcal{A}^i$. The executed profile $\tilde{A}_t = (\tilde{a}_t^K, \tilde{a}_t^S, \tilde{a}_t^F)$ is the observable variable that enters the public record, while the strategic intention remains latent. Given $a_t^{i*} \in \mathcal{A}^i$ and the state vector X_t, execution is determined by

$$\tilde{a}_t^i \sim \mathbb{P}_{\mathrm{I},i}\big(\tilde{a}_t^i \mid a_t^{i*}, X_t\big),$$

and, for each agent $i \in \{K, S, F\}$ and action $a \in \mathcal{A}^i$, it is specified as

$$\mathbb{P}_{\mathrm{I},i}\big(\tilde{a}_t^i = a \mid a_t^{i*}, X_t\big) = \frac{\exp\big(\mathbf{1}\{a = a_t^{i*}\}/T_t\big)}{\sum_{a' \in \mathcal{A}^i} \exp\big(\mathbf{1}\{a' = a_t^{i*}\}/T_t\big)}, \tag{4}$$

where $T_t > 0$ in the base specification is the temperature of the system, which measures the implementation friction of the episode rather than the agent's uncertainty about its own type. To capture strategic learning and institutional maturation jointly, define

$$T_t = T_0 \max\Big\{\frac{H(\mu_t)}{H(\mu_0)} e^{-\eta_{\mathrm{cal}} t},\ \underline{c}\Big\}, \tag{5}$$

with $T_0 > 0$ as the initial temperature, $\eta_{\mathrm{cal}} > 0$ the calendar decay rate, $H(\mu_t) = -\sum_\theta \mu_t(\theta) \ln \mu_t(\theta)$ the entropy of Shannon (1948), and $\underline{c} \in [0, 1]$ the floor of structural noise. In the base specification, the logit distribution (4) satisfies dominance, full support, and maximum entropy of Shannon (1948) subject to those constraints; under the characterization theorem of Jaynes (1957), this functional form is the entropic maximizer. Dominance concentrates most of the mass on a_t^{i*}, while every alternative action retains strictly positive probability as implementation friction. Thus, the MDG fulfills a function equivalent, on the operational plane, to the trembling hand of Selten (1975) and the sequential evaluation system of Kreps and Wilson (1982), so that no branch of the game is excluded by construction and beliefs can be updated by Bayes at reached information sets, in line with the consistency of Fudenberg and Tirole (1991b).

The informational component of the noise declines as the system accumulates information, but this institutional maturation is bounded by the structural floor $\underline{c}$, which persists no matter how much the system learns. When $\underline{c} > 0$, even if $H(\mu_t) \to 0$ the probability of executing exactly a_t^{i*} remains strictly less than one, since the floor preserves full support over $\mathcal{A}^i$ and no branch is excluded in the limit. The special case $\underline{c} = 0$ is interpreted as a limit without residual noise that allows full convergence $\mathbb{P}(\tilde{a}_t^i = a_t^{i*} \mid \mathcal{F}_t^{\text{pub}}) \to 1$ when $H(\mu_t) \to 0$, but sacrifices that full support. The base version with $\underline{c} > 0$ aligns with the absolute continuity or "grain of truth" condition of Kalai and Lehrer (1993), according to which rational learning must not assign zero probability to events that can occur.

The physical outcome m_t is an observable categorical variable on the support $\mathcal{M}_{\text{out}}$, where cont represents the continuation of captivity and the four causes of closure are: $j = 1$ payment of the ransom by the family, $j = 2$ death of the victim in captivity, $j = 3$ tactical rescue by the state, and $j = 4$ an exogenous residual closure channel with constant baseline intensity $\lambda_4 > 0$, which groups voluntary release by the captor without formal ransom, logistical fatigue, and administrative truncations of the sample. Conditional on the captor's type and the material state $\mathcal{C}_t(\theta_K)$, its distribution is obtained from the effective intensities $\tilde{\lambda}_j(t \mid \theta_K)$. Denote by

$$Q_t(m \mid \mathcal{C}_t) := \mathbb{P}(m_t = m \mid \mathcal{C}_t)$$

the physical outcome transition kernel, and by G_t the measurable rule that materializes a concrete realization of m_t from that kernel and a uniform draw. The probability that captivity continues at the end of the period is

$$Q_t(\mathrm{cont} \mid \mathcal{C}_t) = \exp\Big(-\sum_{j=1}^{4} \tilde{\lambda}_j(t \mid \theta_K)\,\Delta t\Big),$$

when the episode closes, the cause is drawn with relative weight $\xi_j(t \mid \theta_K) = \tilde{\lambda}_j / \sum_{\ell=1}^{4} \tilde{\lambda}_\ell$, so that causes with higher intensity concentrate the highest probability of materializing. The physical outcome m_t and the detection indicator d_t form $Y_t = (m_t, d_t) \in \mathcal{M}_{\text{out}} \times \{0,1\}$, the projection of ζ_t onto its last two coordinates. This reduction does not restrict the state's learning, since μ_t is updated with the complete ζ_t, where (α_t^*, γ_t^*) and $\tilde{A}_t$ enter via the implementation likelihood and the material state $\mathcal{C}_t$. Its function is to isolate the stochastic draw of the competing risks block, whose central objects, intensities $\tilde{\lambda}_j$, *hazards* $\bar{h}_j(t)$, and causal mixture $\xi_j(t)$, are fully determined, conditional on $\mathcal{C}_t$, by the daily law of Y_t.

The modeling of risk intensities starts from distinguishing between the instantaneous intensity and its daily probabilistic projection, in line with the proportional hazards tradition of Cox (1972). The cause-specific intensity $\lambda_j(t)$ measures the marginal propensity to close due to cause j as a rate of occurrence per unit of time; thus, it can take any positive value and is not bounded by one. The discrete hazard $\bar{h}_j(t) \in [0,1]$, on the other hand, is the probability that event j materializes during interval t conditional on the system having survived to the beginning of the period. The bar in $\bar{h}_j(t)$ distinguishes this marginal hazard from the public history h_t. Both objects depend on the vector of executed actions $\tilde{A}_t$ generated by the MDG process and not on the latent strategic intentions; the state's continuous instruments (α_t^*, γ_t^*) modify them through the material state $\mathcal{C}_t(\theta_K)$.

For each cause $j \in \{1,2,3\}$, the cause-specific intensity adopts a proportional hazards specification conditioned on the material state $\mathcal{C}_t(\theta_K)$:

$$\lambda_j(t \mid \mathcal{C}_t) = \lambda_{j0}(t) \exp(\mathbf{x}_t' \beta_j), \tag{6}$$

where $\lambda_{j0}(t) \geq 0$ is the baseline risk of cause j, $\mathbf{x}_t$ is a design vector encoding the state and action space $(\theta_F, \theta_V, \theta_S, \theta_K, z, \tilde{A}_t, \alpha_t^*, \gamma_t^*, p_{\text{det},t})$, and β_j contains the latent type-specific effect, so that each type θ_K has its own coefficient $\beta_{K,j}(\theta_K)$ that captures contrasts in criminal technology across organizations. The termination dynamics decompose into four channels: the three strategic causes $j \in \{1, 2, 3\}$ linked to institutional design and a fourth intensity $\lambda_4 > 0$ of exogenous residual closure that groups voluntary release by the captor without formal ransom, logistical fatigue, and administrative truncations of the sample; its intensity is constant baseline and invariant to type θ_K, so its likelihood factorizes orthogonally from the structural parameters in the sense of Lancaster (1990) and Abbring and Berg (2003). The maturation filter $M(t) = \min\{1, (t/T_{\text{mad}})^2\} \in [0, 1]$ scales the strategic intensities to isolate the logistical inertia of the first days of captivity. In discrete time, probabilistic closure cannot be obtained by summing individual transformations $1 - \exp(-\lambda_j)$ because that sum may exceed unity; closure is obtained by first aggregating the intensities and then projecting period survival through the aggregated exponential, in line with Sueyoshi (1992) and Wooldridge (2010). The effective intensities are:

$$\tilde{\lambda}_j(t \mid \mathcal{C}_t) = M(t)\lambda_j(t \mid \mathcal{C}_t) \; (j = 1, 2, 3), \qquad \tilde{\lambda}_4(t) = \lambda_4, \qquad \Delta t = 1. \tag{7}$$

The continuation probability, the daily closure rate, and the conditional distribution of causes are obtained from:

$$p_{\text{Cont},t} = \exp\Big(-\sum_{j=1}^{4} \tilde{\lambda}_j(t)\Big), \tag{8}$$

$$q(t) = 1 - p_{\text{Cont},t}, \tag{9}$$

$$\xi_j(t) = \frac{\tilde{\lambda}_j(t)}{\sum_{\ell=1}^{4} \tilde{\lambda}_\ell(t)}, \quad j = 1, \ldots, 4, \tag{10}$$

so that $\bar{h}_j(t) = q(t)\xi_j(t)$ is the marginal hazard of cause j and $\sum_j \bar{h}_j(t) = q(t) \leq 1$ by construction. The probability $\xi_j(t)$ acts as the market share of each risk and orthogonalizes two dimensions of the process: the governance of duration (the when), determined by the aggregate density of the intensities, and the governance of the nature of the closure (the how), governed by the relative weights. It is through $\xi_j(t)$ that Bayesian learning exerts its strategic role: as μ_t concentrates on the true type, the probability mass is redistributed among exit causes, shifting the likelihood toward the outcomes that are more probable under that type, without altering the temporal scale of captivity. The probability of detecting collusion $p_{\mathrm{det},t}$ is specified through a logistic response function conditioned on the state's instruments:

$$p_{\mathrm{det},t}(\theta_K) = \Lambda(\eta_0(\theta_K) + \eta_1\alpha_t^* + \eta_2\gamma_t^*), \qquad \Lambda(u) = \frac{1}{1+\exp(-u)}. \tag{11}$$

where $\eta_0(\theta_K)$ is the type-specific intercept that captures the baseline detectability of the criminal organization. Because different organizations differ in their propensity to use traceable formal financial channels, d_t is informative about θ_K. The parameters $\eta_1 > 0$ and $\eta_2 > 0$ capture the marginal effect of the financial block and operational pressure. The logistic function guarantees $p_{\mathrm{det},t}(\theta_K) \in (0,1)$ and monotonicity increasing in (α_t^*, γ_t^*), so that greater financial interdiction or operational pressure raises the probability of observing $d_t = 1$, with a baseline level that varies according to the type's concealment technology.

To operationalize the informative quality in rescue planning, the precision index ι_t is defined as the probability mass of the modal type,

$$\iota_t = \max_{\theta\in\Theta_K} \mu_t(\theta), \qquad \hat{\theta}_t = \arg\max_{\theta\in\Theta_K} \mu_t(\theta), \tag{12}$$

where $\hat{\theta}_t$ is the captor's estimated profile. Let $s_t \in \{0,1\}$ be the binary indicator of victim survival at the end of the operational margin of the period, where $s_t = 1$ means that the victim survives and $s_t = 0$ that she does not survive; this indicator operates in the rescue branch and does not redefine the categorical support of m_t. Conditional on a focal rescue attempt, the scalar probability of survival $p_{\text{surv},t}$ is modeled through a logistic response that penalizes the misalignment between the estimated profile and the true type:

$$p_{\text{surv},t}(\iota_t, \hat{\theta}_t, \theta_K) := \Lambda\big(\alpha_{\text{leth}}(\theta_K) + \beta_R \cdot \iota_t \cdot \mathbf{1}\{\hat{\theta}_t = \theta_K\}\big), \tag{13}$$

where $\alpha_{\text{leth}}(\theta_K)$ is the intrinsic lethality of the type, $\beta_R > 0$ is the productivity of accumulated intelligence, and $\mathbf{1}\{\hat{\theta}_t = \theta_K\}$ indicates whether the state has correctly identified the captor. When the identification is correct, ι_t enhances $p_{\text{surv},t}$ according to β_R; when there is an identification error, the accumulated intelligence becomes inoperative, and the survival probability collapses to the baseline level determined by $\alpha_{\text{leth}}(\theta_K)$.

The daily decision mechanism is projected onto a longitudinal framework via the episode survival function $S(t)$ and the cumulative incidence functions $F_j(t)$. $S(t)$ denotes the probability that the episode remains active until the end of period t, and $F_j(t)$ measures the cumulative probability of closure by cause j up to that same horizon. With $S(0) = 1$, these objects are constructed as

$$S(t) = \prod_{t'=1}^{t} p_{\text{Cont},t'}, \qquad F_j(t) = \sum_{t'=1}^{t} \bar{h}_j(t')\, S(t'-1), \qquad S(t) + \sum_{j=1}^{4} F_j(t) = 1. \tag{14}$$

The constraint $S(t) + \sum_j F_j(t) = 1$ guarantees the accounting consistency of the system, since all probability not assigned to the continuation of the episode is distributed among the four causes of closure. By weighting each marginal hazard $\bar{h}_j(t')$ by the remaining survival $S(t'-1)$, the system operates under the formal treatment of competing risks and operationalizes the state's strategic dilemma between intervening immediately or delaying action to accumulate informational capital ι_t.

The technical probability of capture is modeled via a logistic response that keeps the probability within the unit interval and allows public instruments to smoothly modify the marginal effectiveness of the intervention:

$$p_{\text{cap}}(a, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*) = \Lambda\big(\delta_a + c_0(\theta_i) + c_\alpha(\theta_i)\alpha_t^* + c_\gamma(\theta_i)\gamma_t^* + c_S(\theta_S)\big), \quad (15)$$

where δ_a captures the impact of the executed intervention mode itself, $c_0(\theta_i)$ allows for baseline heterogeneity by type, $c_\alpha(\theta_i)$ and $c_\gamma(\theta_i)$ capture how the financial block and operational pressure alter the probability of capture against that specific type, and $c_S(\theta_S)$ incorporates the institutional capacity of the state. The effective probability of capture is defined as the expectation of p_{cap} over the stochastic realizations of the executed state action, conditional on the tuple $\mathcal{Q}_t^{\text{Cap}} := (a_t^{S*}, \alpha_t^*, \gamma_t^*, \theta_K, \theta_S)$:

$$\begin{aligned} \tilde{p}_{\text{cap},t}(\theta_i) &:= \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}}\big[p_{\text{cap}}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*)\big] \\ &= \sum_{a \in \mathcal{A}^S} \mathbb{P}_{\text{I},S}(a \mid a_t^{S*}, X_t) \cdot p_{\text{cap}}(a, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*), \end{aligned} \quad (16)$$

where $\mathcal{Q}_t^{\text{Cap}}$ collects the optimal intention of the state, the executed instruments, and the relevant types of the episode. This formulation separates two levels: $\mathbb{P}_{\text{I},S}$ describes the probability that the state's intention is translated into a concrete action through the MDG process, and p_{cap} measures the technical probability of capture under the instruments actually executed; in this way, $\tilde{p}_{\text{cap},t}(\theta_i)$ combines implementation uncertainty and operational capacity, avoiding mechanically identifying the planned decision with the observable police outcome.

The mechanism obtains additional information through acoustic evidence at the micro level and learning by absence of communication. In each period t, the state can observe a voice window or face operational silence. When there is a signal, it is summarized in a feature vector $x_t^{obs} \in \mathbb{R}^k$ that decomposes the observed signal into the true acoustic pattern of the type ($x_t^{true}(\theta_K)$), channel noise (ε_L), and strategic camouflage (ε_S), with both perturbations being Gaussian and independent of each other:

$$x_t^{obs} = x_t^{true}(\theta_K) + \varepsilon_L + \varepsilon_S, \qquad \varepsilon_L \sim \mathcal{N}(0, \Sigma_L), \quad \varepsilon_S \sim \mathcal{N}(0, \Sigma_S), \quad \varepsilon_L \perp \varepsilon_S. \tag{17}$$

The Bayesian architecture of the mechanism rests on four likelihoods that convert each observed day into evidence about the captor's type θ_K. The first captures the information contained in executed actions as realizations of the MDG process ($\mathcal{L}_{I,t}$); the second and third collect the information from the physical outcome m_t and the detection signal d_t through the competing risks block; and the fourth incorporates acoustic evidence or operational silence through the communication likelihood $\mathcal{L}_{C,t}$. Analogously to the job search models of McCall (1970), a continuation day updates the posterior by non-occurrence and a closure updates it through its cause, so that the model converts each observable realization into statistical information without requiring an additional observational layer. The implementation likelihood is the first of these four sources:

$$\mathcal{L}_{I,t}(\theta_K) := \prod_{i \in \{F,K,S\}} \mathbb{P}_{\mathrm{I},i}\big(\tilde{a}_t^i \mid a_t^{i*}(\theta_K), X_t\big), \tag{18}$$

where a_t^{K*} may depend directly on θ_K, while a_t^{F*} and a_t^{S*} depend on the public history and on μ_t; if an executed action was unlikely under the plan induced by a type, that type receives lower posterior weight.

The likelihood of the physical outcome m_t distinguishes two regimes. On a continuation day no competing risk materializes and the absence of closure favors types whose exit rates predict greater continuity:

$$\mathcal{L}_H^{\text{cont}}(t \mid \theta_K) = p_{\text{Cont},t|\theta_K} = \exp\Big(-\sum_{j=1}^{4} \tilde{\lambda}_j(t \mid \theta_K)\Delta t\Big). \tag{19}$$

When the day ends due to cause $j \in \{1,2,3,4\}$, the outcome identifies the type by the cause that produced it:

$$\mathcal{L}_H^{\text{out}}(t, j \mid \theta_K) = \bar{h}_j(t \mid \theta_K). \tag{20}$$

Let $g : \{1,2,3,4\} \to \mathcal{M}_{\text{term}}$ be the bijection $g(1) = \text{pay}$, $g(2) = \text{kill}$, $g(3) = \text{res}$, $g(4) = \text{rel}$. Both regimes are written as a single expression:

$$\mathbb{P}(m_t \mid \theta_K, \mathcal{C}_t) = p_{\text{Cont},t|\theta_K}^{\mathbf{1}\{m_t=\text{cont}\}} \prod_{j=1}^{4} \bar{h}_j(t \mid \theta_K)^{\mathbf{1}\{m_t=g(j)\}}, \tag{21}$$

a five-cell multinomial (cont and $\mathcal{M}_{\text{term}}$) where g converts index j into the label of m_t.

Regardless of the physical outcome, the detection signal d_t provides a second source of information with Bernoulli likelihood:

$$\mathbb{P}(d_t \mid \theta_K, \alpha_t^*, \gamma_t^*) = p_{\text{det},t}(\theta_K)^{d_t}\big(1 - p_{\text{det},t}(\theta_K)\big)^{1-d_t}. \tag{22}$$

The mechanism incorporates a third source of information through micro-level acoustic evidence and learning by absence of communication. Each type θ_K possesses a reference profile $(\bar{x}(\theta_K), \Sigma(\theta_K))$; the diagonal Gaussian likelihood assigns greater weight to types whose profiles make the observed signal more plausible:

$$\mathcal{L}_{\text{voz},t}(\theta_K) \propto \exp\left(-\frac{1}{2}\sum_{i=1}^{k} \frac{\big(x_{t,i}^{obs} - \bar{x}_i(\theta_K)\big)^2}{\tilde{\sigma}_i(\theta_K)^2}\right). \tag{23}$$

Let $V_t \in \{0,1\}$ be the indicator of voice emission at t and $\pi_{\text{call}}(\theta_K) \in (0,1)$ the contact probability of each type, interpreted as logistical urgency; both the call and silence are informative. The type-conditional communication likelihood is

$$\mathcal{L}_{C,t}(\theta_K \mid V_t) = \begin{cases} \left[\mathcal{L}_{\text{voz},t}(\theta_K)\,\pi_{\text{call}}(\theta_K)\right]^{\omega_{\text{voz}}}, & V_t = 1, \\ \left[1 - \pi_{\text{call}}(\theta_K)\right]^{\omega_{\text{voz}}}, & V_t = 0, \end{cases} \tag{24}$$

where $\omega_{\text{voz}} \in [0,1]$ is the learning weight; when there is no active information, $\omega_{\text{voz}} = 0$ and $\mathcal{L}_{C,t} \equiv 1$.

The three sources, executed actions, physical outcome, and detection signal, come from distinct operational margins connected by the same material state, which allows factoring the joint likelihood of the physical-institutional block:

$$\mathcal{L}_{F,t}(\theta_K) = \underbrace{\mathcal{L}_{I,t}(\theta_K)}_{\text{executed actions}} \cdot \underbrace{\mathbb{P}(m_t \mid \theta_K, \mathcal{C}_t)}_{\text{competing risks}} \cdot \underbrace{\mathbb{P}(d_t \mid \theta_K, \alpha_t^*, \gamma_t^*)}_{\text{detection}}. \tag{25}$$

If continuity and positive detection are observed, the risks block reduces the weight of types with high hazards and the detection block reduces that of types with low probability of fiscalization; if rescue and absence of detection are observed, the competitive core favors types with higher rescue hazard; $\mathcal{L}_{I,t}$ penalizes types for which the executed triplet was improbable under the MDG. The unified Bayes rule combines the physical-institutional and communication blocks:

$$\mu_{t+1}(\theta_K) = \frac{\mu_t(\theta_K)\,\mathcal{L}_{F,t}(\theta_K)\,\mathcal{L}_{C,t}(\theta_K \mid V_t)}{\sum_{\tilde{\theta}\in\Theta_K} \mu_t(\tilde{\theta})\,\mathcal{L}_{F,t}(\tilde{\theta})\,\mathcal{L}_{C,t}(\tilde{\theta} \mid V_t)}. \tag{26}$$

The degenerate rule illustrates the modularity of the system. When $\omega_{\text{voz}} = 0$ or there is no voice data, $\mathcal{L}_{C,t} \equiv 1$ and updating relies only on the physical-institutional record $(\tilde{A}_t, m_t, d_t)$; if implementation evidence is additionally set aside by fixing $\mathcal{L}_{I,t} \equiv 1$, updating reduces to the minimal record (m_t, d_t). The principle governing which objects enter the posterior is simple, since the Bayes operator can update only on realizations publicly observable at the close of the period. Therefore, strategic intentions a_t^{i*} do not enter directly; only their observable realizations $\tilde{A}_t$ do so through $\mathcal{L}_{I,t}$; and capture probabilities $\tilde{p}_{\text{cap},t}$ and survival probabilities $p_{\text{surv},t}$ are already contained in the law of the physical outcome $\mathbb{P}(m_t \mid \theta_K, \mathcal{C}_t)$, so incorporating them separately would imply double counting. Information about θ_K is not introduced as an exogenous signal but as an observable product of a strategic optimization process subjected to the MDG filter, so that, under learning with increasing information in the sense of Blackwell and Dubins (1962), the sequence of realizations progressively filters incompatible types.

2.4. *Effective probabilities*

In all decision branches, the probabilities that weight the outcomes are defined as conditional expectations over the realizations of the MDG process induced by the conditioning tuple $\mathcal{Q}_t$ pertinent to each action. This separation between the underlying technological probability and its stochastic realization avoids mechanically identifying the strategic intention with the observable result. The four effective probabilities that feed the Bellman objects are:

$$\tilde{p}_{\text{cap},t}(\theta_i) := \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}}\big[p_{\text{cap}}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*)\big], \tag{27}$$

$$\tilde{p}_{\text{pay},t}(\theta_i) := \mathbb{E}_{\tilde{A}_t \mid \mathcal{Q}_t^{\text{Cont}}}\big[\mathbb{P}(m_t = \text{pay} \mid \tilde{A}_t, X_t', \theta_K = \theta_i)\big], \tag{28}$$

$$\tilde{p}_{\text{surv},t}(\theta_i) := \mathbb{E}_{\tilde{A}_t \mid \mathcal{Q}_t^{\text{Coop}}}\big[p_{\text{surv},t}(\iota_t, \hat{\theta}_t, \theta_K = \theta_i)\big], \tag{29}$$

$$\tilde{p}_{\text{rel},t}(\theta_i) := \mathbb{E}_{\tilde{A}_t \mid \mathcal{Q}_t^{\text{Col}}}\big[\mathbb{P}(m_t = \text{rel} \mid \tilde{A}_t, R, \theta_K = \theta_i)\big], \tag{30}$$

The conditioning tuples specify the branch action and the executed instruments for each expectation. For capture, $\mathcal{Q}_t^{\text{Cap}} := (a_t^{S*}, \alpha_t^*, \gamma_t^*, \theta_K, \theta_S)$. The three remaining branches share the structure $\mathcal{Q}_t^b := (a_t^{S*}, a_t^{b*}, a_b, \alpha_t^*, \gamma_t^*, \mu_t, \theta_K)$, where $b \in \{\mathrm{Cont}, \mathrm{Coop}, \mathrm{Col}\}$ and a_b is the corresponding branch action (a_{cont}, a_{coop}, or a_{col}). In this way, the model maintains coherence between the chosen strategic branch and the evaluated stochastic realization.

2.5. Rational behavior and best responses

Captor's behavior.

In each period t, the captor solves an optimal stopping problem, selecting the action $a_t^K \in \mathcal{A}^K = \{a_{\mathrm{rel}}, a_{\mathrm{kill}}, a_{\mathrm{cont}}\}$ that maximizes its value function given its private type θ_K, the public history h_t, and the institutional environment θ_S. Comparing the utilities of releasing, killing, and continuing captivity, its best response in compact form is

$$V_t^K(\theta_K) = \max\{U_{\mathrm{rel},t}^K(\theta_K), U_{\mathrm{kill},t}^K(\theta_K), V_{\mathrm{cont},t}^K(\theta_K)\}. \tag{31}$$

In the death branch, the payoff weights the reputational benefit $\eta(\theta_K)$ against the expected penalty for capture:

$$U_{\mathrm{kill}}^K(\theta_K, \theta_S) = \big(1 - \tilde{p}_{\mathrm{cap},t}(\theta_K)\big)\eta(\theta_K) - \tilde{p}_{\mathrm{cap},t}(\theta_K)\, F_{\mathrm{cap}}(\theta_K, \theta_S). \tag{32}$$

In the release branch, the captor incurs the disutility of yielding without obtaining rent, reflected in lost reputation or foregone income:

$$U_{\mathrm{rel}}^K(\theta_K) = -\kappa_{\mathrm{rel}}(\theta_K), \qquad \kappa_{\mathrm{rel}}(\theta_K) \geq 0. \tag{33}$$

Continuation combines expected ransom net of financial blockade, operating cost, expected capture penalty, and discounted future value by type. Since θ_K is private information of the captor, μ_t is not an uncertainty for it but a strategic object, since the evolution of μ_{t+1} conditions the state's future policy and forms part of its intertemporal calculation.

$$
\begin{aligned}
V^K_{\text{cont},t}(\theta_K) = \tilde{p}_{\text{pay},t}(\theta_K)\, R(1-\alpha_t^*) - C_t(\gamma_t^*, \theta_K) \\
- \tilde{p}_{\text{cap},t}(\theta_K)\, F_{\text{cap}}(\theta_K, \theta_S) \\
+ \tilde{\beta}(\theta_K)\big(1 - \tilde{p}_{\text{cap},t}(\theta_K)\big)\, \mathbb{E}\big[V^K_{t+1}(\mu_{t+1}(a_{\text{cont}}))\big]\,,
\end{aligned}
\tag{34}
$$

where $C_t(\gamma_t^*, \theta_K)$ is a convex cost function that is increasing in γ_t^* and varies asymmetrically across types, generating the single-crossing property that renders the persistence of captivity an informative signal for the state's Bayesian updating; $\tilde{\beta}(\theta_K) \in (0,1)$ is the type-specific discount factor; $\tilde{p}_{\text{pay},t}$ and $\tilde{p}_{\text{cap},t}$ are the effective probabilities from Section 2.4; and the continuation of future value is taken over the evolution of the belief μ_{t+1} induced by the action a_{cont}. The optimal action of the captor a_t^{K*} arises from comparing the three branches:

$$
a_t^{K*} = \arg\max_{a_t^K \in \mathcal{A}^K} \mathcal{U}_t^K(a_t^K \mid h_t, \theta_K, \theta_S), \quad \mathcal{U}_t^K = \begin{cases} U^K_{\text{rel}}(\theta_K), & \text{if } a_t^K = a_{\text{rel}}, \\ U^K_{\text{kill}}(\theta_K, \theta_S), & \text{if } a_t^K = a_{\text{kill}}, \\ V^K_{\text{cont},t}(\theta_K), & \text{if } a_t^K = a_{\text{cont}}. \end{cases}
\tag{35}
$$

Family's behavior.

In each period t, the family chooses $a_t^F \in \mathcal{A}^F = \{a_{\text{coop}}, a_{\text{col}}\}$ by maximizing its expected utility over the uncertainty regarding the captor's type, weighted by its private information set $\mathcal{I}_t^F$. In the cooperation branch, the family follows the institutional path, paying the cooperation effort cost $e_t(\gamma_t, \theta_F)$, which captures the frictions and risk of retaliation under interdiction, and is convex and increasing in γ_t with asymmetric variation across family types that generates a single-crossing property analogous to that of the captor; in return, it obtains the expected value of the victim's survival weighted by $V_L > 0$:

$$\mathcal{U}_t^F(a_{\text{coop}} \mid \theta_F) = \mathbb{E}_{\theta_K \mid \mathcal{I}_t^F}\big[\tilde{p}_{\text{surv},t}(\theta_K)\big]\, V_L - e_t(\gamma_t, \theta_F). \tag{36}$$

In the collusion branch, the family clandestinely pays the ransom R to secure the victim's release, but discounts the probability of detection by the state and the fine F_{col}:

$$\mathcal{U}_t^F(a_{\text{col}} \mid \theta_F) = \mathbb{E}_{\theta_K \mid \mathcal{I}_t^F}\big[\tilde{p}_{\text{rel},t}(\theta_K)\big]\, V_L - R - p_{\text{det},t}\, F_{\text{col}}, \tag{37}$$

where $\tilde{p}_{\text{surv},t}$ and $\tilde{p}_{\text{rel},t}$ are the effective probabilities from Section 2.4. The optimal action of the family arises from comparing the two branches:

$$a_t^{F*} = \arg\max_{a_t^F \in \mathcal{A}^F} \mathcal{U}_t^F(a_t^F \mid \theta_F), \qquad \mathcal{U}_t^F = \begin{cases} \mathcal{U}_t^F(a_{\text{coop}}), & \text{if } a_t^F = a_{\text{coop}}, \\ \mathcal{U}_t^F(a_{\text{col}}), & \text{if } a_t^F = a_{\text{col}}. \end{cases} \tag{38}$$

State's behavior.

In each period t, given the public history $\mathcal{F}_t^{\text{pub}}$ and the belief vector $\mu_t \in \Delta(\Theta_K)$, the state solves a Bayesian dual control problem by choosing the discrete action $a_t^S \in \{a_{\text{res}}, a_{\text{neg}}\}$ and the continuous instruments $(\alpha_t, \gamma_t) \in [0,1]^2$. Its objective is not to maximize its own utility but to minimize an expected social loss that combines human risk, economic costs, operational costs, and an informational motive. The conditional loss by type varies by branch:

$$L_t(a_t^S, \alpha_t, \gamma_t, \theta_K, \iota_t) = \begin{cases} V_t^R(\iota_t, \hat{\theta}_t, \theta_K, \alpha_t, \gamma_t), & \text{if } a_t^S = a_{\text{res}}, \\ V_t^N(\theta_K, \alpha_t, \gamma_t), & \text{if } a_t^S = a_{\text{neg}}, \end{cases} \tag{39}$$

where under tactical rescue the state incurs the cost of a lethal failure of the operation and the cost of deployment, $V_t^R := \omega_k[1 - p_{\text{surv},t}(\iota_t, \hat{\theta}_t, \theta_K)] + C_{\text{ops}}(\gamma_t, \alpha_t; \theta_K)$, and under negotiation it incurs the cost of net criminal transfer, the lethality of captivity, and the maintenance of the containment ring, $V_t^N := \omega_p R(1 - \alpha_t) + \omega_k \bar{h}_2(t \mid$

$\theta_K, \mathcal{C}_t) + C_{\text{maint}}(\gamma_t, \alpha_t; \theta_K)$; here $\bar{h}_2(t \mid \theta_K, \mathcal{C}_t) = q(t \mid \theta_K)\, \xi_2(t \mid \theta_K)$ is the marginal hazard of death from captivity, derived from the competing risks system (10), and the weights $\omega_k > 0$ and $\omega_p > 0$ calibrate the human loss and economic transfer respectively. The cost functions C_{ops} and C_{maint} are quadratic, continuous, and convex in (α_t, γ_t), with parameters that depend on θ_K to capture logistical, territorial, and financial heterogeneity; the cross terms in $\gamma_t\alpha_t$ capture complementarity or substitutability between financial block and operational pressure depending on the type of criminal organization faced. The state's executable problem is

$$
\begin{aligned}
(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t)} \Big\{ \sum_{\theta \in \Theta_K} \mu_t(\theta)\, L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) \\
+ \Pi^S_{t, a^S}(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\},
\end{aligned}
\tag{40}
$$

where $\Gamma_t(\mu_t)$ incorporates the belief-weighted constraints IC^K, IR^K, and IR^F. To prevent the state from choosing arbitrarily extreme instruments under uncertainty, type- and branch-specific counterfactual benchmarks are constructed to indicate what the state would do if it knew θ with certainty: $(\alpha^{\text{bench}}_{t,b}(\theta), \gamma^{\text{bench}}_{t,b}(\theta)) \in \arg\min_{(\alpha,\gamma)} V_t^b(\theta, \alpha, \gamma)$ for $b \in \{R, N\}$. The Bayesian center of branch b weights these benchmarks by the current belief:

$$
(\alpha^{\mu}_{t,b}, \gamma^{\mu}_{t,b}) := \sum_{\theta \in \Theta_K} \mu_t(\theta)(\alpha^{\text{bench}}_{t,b}(\theta), \gamma^{\text{bench}}_{t,b}(\theta)),
$$

and the penalty $\Pi^S_{t,b}$ measures how much the candidate policy deviates from this center:

$$
\Pi^S_{t,b}(\alpha_t, \gamma_t; \mu_t) = \chi_\alpha(\alpha_t - \alpha^{\mu}_{t,b})^2 + \chi_\gamma(\gamma_t - \gamma^{\mu}_{t,b})^2, \qquad \chi_\alpha, \chi_\gamma \geq 0, \tag{41}
$$

which allows exploring alternative instruments but charges a premium when the instrument moves away from the Bayesian center. The informational motive

$-\psi_H \Delta H_t$ operationalizes active exploration, as the state recognizes that (α_t, γ_t) modifies risk intensities and the probability of detection, so a policy may be valuable not only for reducing current material loss but for separating likelihoods and reducing future uncertainty, in line with the dynamic acquisition of information of Zhong (2022). The expected information gain is the expected reduction in the entropy of Shannon (1948) over the joint public signal (m_t, d_t),

$$\Delta H_t(\alpha_t, \gamma_t) = H(\mu_t) - \sum_m \sum_{d \in \{0,1\}} \mathbb{P}(m_t = m, d_t = d \mid \mu_t, \alpha_t, \gamma_t, \mathcal{C}_t)\, H(\mu_{t+1}^{m,d}), \tag{42}$$

where $\mu_{t+1}^{m,d}$ is the counterfactual posterior that would result from observing the pair (m, d) under the candidate policy, and the joint probability factorizes by conditional independence given the type: $\mathbb{P}(m, d \mid \mu_t, \alpha_t, \gamma_t, \mathcal{C}_t) = \sum_\theta \mu_t(\theta)\, \mathbb{P}(m_t = m \mid \theta, \mathcal{C}_t)\, \mathbb{P}(d_t = d \mid \theta, \alpha_t, \gamma_t)$. The voice channel V_t and the implementation signal $\tilde{A}_t$ are excluded from ΔH_t, since the former does not depend on (α_t, γ_t) and is modeled separately in $\mathcal{L}_{C,t}$; the latter captures implementation friction, not type discrimination. If $\psi_H = 0$, the state is purely myopic; if $\psi_H > 0$, policies that buy learning receive a discount on the loss; when μ_t concentrates on the true type, ΔH_t falls and endogenously shuts down the exploratory motive. The solution is obtained by partition: first, the continuous instruments are minimized within each branch, incorporating the penalty and the entropic motive, and then the optimal floors are compared. The loss floors of each branch are

$$\widetilde{V}_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t) = \min_{(\alpha_t,\gamma_t)\in\Gamma_t^R(\mu_t)} \Big\{ \sum_{\theta\in\Theta_K} \mu_t(\theta)\, V_t^R(\iota_t, \hat{\theta}_t, \theta, \alpha_t, \gamma_t) + \Pi_{t,R}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\}, \tag{43}$$

$$\widetilde{V}_t^{N,*}(\mu_t) = \min_{(\alpha_t,\gamma_t)\in\Gamma_t^N(\mu_t)} \Big\{ \sum_{\theta\in\Theta_K} \mu_t(\theta)\, V_t^N(\theta, \alpha_t, \gamma_t) + \Pi_{t,N}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\}. \tag{44}$$

so that each floor integrates the expected material loss, the penalty for deviation from the Bayesian center, and the informational subsidy. The state's optimal social cost is

$$\mathcal{L}_t^{S*} = \min\big\{ \widetilde{V}_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t),\ \widetilde{V}_t^{N,*}(\mu_t) \big\}. \tag{45}$$

If $(\alpha^*_{\text{res}}, \gamma^*_{\text{res}})$ and $(\alpha^*_{\text{neg}}, \gamma^*_{\text{neg}})$ denote the continuous argminima of the rescue and negotiation branches, the optimal executed policy is

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) = \begin{cases} (a_{\text{res}}, \alpha^*_{\text{res}}, \gamma^*_{\text{res}}), & \text{if } \widetilde{V}_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t) \le \widetilde{V}_t^{N,*}(\mu_t), \\ (a_{\text{neg}}, \alpha^*_{\text{neg}}, \gamma^*_{\text{neg}}), & \text{if } \widetilde{V}_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t) > \widetilde{V}_t^{N,*}(\mu_t). \end{cases} \tag{46}$$

In case of numerical equivalence between the loss floors, rescue prevails due to the weak inequality of the first condition.

Individual rationality and incentive compatibility (IR/IC).

The minimization of the state's social loss is bounded by three design constraints that guarantee the viability of learning, the deterrence of crime, and civil cooperation. Since the state operates under incomplete information, IC^K and IR^K are formulated as expected-value constraints under the posterior belief μ_t, in line with Bayesian mechanism design of Holmström and Myerson (1983). This formulation does not require mechanical type-by-type separation in every period, but imposes the constraint on the prevailing type distribution; it converges to the standard condition for the true type when $\mu_t \to \delta_{\theta_K^*}$.

First, the *Bayesian incentive compatibility of the kidnapper* (IC^K) demands that, in expected value over the true type, the trajectory assigned to each type dominates the mimicry of any rival type:

$$\mathbb{E}_{\theta\sim\mu_t}\big[V_t^K(a^*(\theta) \mid \theta, \alpha_t, \gamma_t) - V_t^K(a^*(\theta_j) \mid \theta, \alpha_t, \gamma_t)\big] \geq 0, \qquad \forall\, \theta_j \in \Theta_K. \tag{47}$$

When $\mu_t(\theta^*) \to 1$, the constraint collapses into the incentive compatibility of the true type against each rival mimicry.

The second is the *Bayesian participation constraint of the kidnapper* (IR^K), which operates under a logic of strategic deterrence: the reservation utility linked to peaceful release, $U_{\text{rel}}^K(\theta) = -\kappa_{\text{rel}}(\theta)$, must dominate in expected value the best criminal alternative between continuation and executing the hostage:

$$\mathbb{E}_{\theta\sim\mu_t}\Big[U_{\text{rel}}^K(\theta) - \max\big\{V_{\text{cont},t}^K(\theta, \alpha_t, \gamma_t), U_{\text{kill}}^K(\theta, \theta_S)\big\}\Big] \geq 0. \tag{48}$$

The third is the *participation constraint of the family* (IR^F), which disciplines the financing channel by requiring that the expected utility of cooperating with the state dominates the return of colluding via clandestine payment of the ransom:

$$\mathcal{U}_t^F(a_{\text{coop}}) \geq \mathcal{U}_t^F(a_{\text{col}}). \tag{49}$$

The three constraints jointly define the feasible set $\Gamma_t(\mu_t)$ within which the state optimizes, and anchor the functional form of the likelihood by restricting the space of admissible strategic profiles in PBE.

2.6. Type Identification Theorem for Rational Types in Unknown Environments

Bayesian convergence does not imply identification. As Doob (1953) and Blackwell and Dubins (1962) show, beliefs can stabilize without concentrating on the true type when different profiles generate observationally equivalent signals. In the sense of Kalai and Lehrer (1993), Bayesian learning may suffice to predict the future path correctly and stop at a subjective equilibrium; analogously, the self-confirming equilibrium of Fudenberg and Levine (1993) allows beliefs to be correct on the observed trajectory and wrong on unvisited contingencies. Therefore, predictive coherence on the path does not necessarily identify the true type. This section establishes when the public history of the mechanism discriminates types in the limit, combining Bayesian coherence, statistical separation between signal laws, and the singularity criterion of Kakutani (1948). The strategy requires three ingredients: a public filtration over which μ_t is a martingale; a long-run identification condition that prevents distinct types from explaining the same observable trajectory; and uniform separation in total variation provided by heterogeneity in $\beta_{K,j}(\theta_K)$, which induces mutual singularity of trajectory laws and collapse of posterior mass on false types.

The public signal ζ_t is the central object that feeds the filtration. Once the PBE is fixed, $\mathbb{P}_{\mathrm{E}}$ is the ex-ante law on Ω that describes the entire experiment before knowing the captor's type; the type is realized according to the structural prior $\theta_K \sim \mu_0$, and conditional on this type, the public policy, executed actions, outcomes, and public history h_t unfold. For each counterfactual type $\theta \in \Theta_K$, $\mathbb{P}_{\mathrm{C},\theta}$ denotes the corresponding type-conditional law on Ω under the same fixed PBE.

Definition 1 (Public signal and filtration)**.** (i) Public signal ζ_t. The public signal ζ_t is the random vector that summarizes what is observable at the end of period t. The order of the components reproduces the way in which the public history incorporates, at the end of each period, the chosen instruments, the executed actions, and the observed results: $\alpha_t \in [0,1]$ and $\gamma_t \in [0,1]$ are the continuous instruments of the state; $(\tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S)$ are the executed discrete actions that appear as public observables after the MDG process and do not replace the equilibrium intentions a_t^{i*} in the record; $m_t \in \mathcal{M}_{\text{out}}$ is the physical outcome ($\mathcal{M}_{\text{out}}$ is the physical outcome space defined in Section 2); and $d_t \in \{0,1\}$ is the detection signal.

(ii) Value space. ζ_t takes values in the product space $\mathcal{Z}_t \subseteq \mathbb{R}^2 \times \mathcal{A}^F \times \mathcal{A}^K \times \mathcal{A}^{S,\text{disc}} \times \mathcal{M}_{\text{out}} \times \{0,1\}$, where the discrete components are symbols in finite sets and the continuous coordinates are real. Thus, ζ_t encodes what is observable at the close of t and summarizes the public innovation that feeds the next history of the episode.

(iii) Public filtration $\{\mathcal{F}_t^{\text{pub}}\}_{t\geq 0}$. The initial instant is fixed by $\mathcal{F}_0^{\text{pub}} := \sigma(z, \theta_V)$, i.e., the σ-algebra generated by the observable geographic location at the opening of the episode. For each integer $t \geq 1$, $\mathcal{F}_t^{\text{pub}} := \sigma(z, \zeta_0, \ldots, \zeta_{t-1})$, i.e., the σ-algebra generated by the initial location and by the signals observed strictly before starting period t, equivalent to “at the opening of day t” in the temporal reading of the document.

(iv) Measurability and measurable structure in $\mathcal{Z}_t$. It is assumed that each $\zeta_t : (\Omega, \mathcal{F}) \to (\mathcal{Z}_t, \mathcal{B}(\mathcal{Z}_t))$ is measurable and that z is observable at the opening of the episode, so that $\sigma(z, \zeta_0, \ldots, \zeta_{t-1})$ is well defined as a sub-σ-algebra of $\mathcal{F}$.

$\square$

Once the public filtration is fixed, each counterfactual type induces its own law on the daily signal. The following definition formalizes this law as a predictive kernel, the object that the Bayes operator reweights in each period and that connects the space of latent types with the space of observables.

Definition 2 (Predictive likelihood under a counterfactual type)**.** Fixing a PBE with profile (χ, s^F, s^K), the stochastic dynamics of the mechanism, and a counterfactual type $\theta \in \Theta_K$, the predictive likelihood of period t is defined by

$$\ell_t(\theta; J) \; := \; \mathbb{P}_{\mathrm{C},\theta}\big(\zeta_t \in J \,\big|\, \mathcal{F}_t^{\mathrm{pub}}\big), \qquad J \in \mathcal{B}(\mathcal{Z}_t), \tag{50}$$

where $\mathcal{Z}_t$ is the space of possible values of the public signal and J represents an observable subset of that space. The conditioning is interpreted as the conditional expectation of $\mathbf{1}\{\zeta_t \in J\}$ given the public information accumulated up to t, defined up to $\mathbb{P}_{\mathrm{C},\theta}$-null sets through standard regular versions. The likelihoods $\mathcal{L}_{F,t}$ and $\mathcal{L}_{C,t}$ are the operational realization of this kernel for the physical block and the communication block of the period.

□

The model architecture postulates that the likelihood of observable trajectories, denoted by $\mathcal{L}_t(\theta) = \mathcal{L}_{F,t}(\theta)\mathcal{L}_{C,t}(\theta)$, is not a statistically invariant object, but is subject to the Lucas Critique (1976); under any modification in the mechanism χ, whether due to the tightening of instruments or a more aggressive operational stance of the state (a_t^S), agents reoptimize their strategies in response to the new institutional environment, thereby recomposing the equilibrium map (χ, s^F, s^K) and the predictive likelihood of the behaviors encoded in the informative stream ζ_t. In this framework, the conditional kernel ℓ_t acts as the formal bridge between the space of latent states ($\theta \in \Theta_K$) and the space of observables ($\mathcal{Z}_t$), where for each hypothetical type θ, the measure $\ell_t(\theta; J)$ quantifies the probability distribution of the daily signal over the Borel sets $J \in \mathcal{B}(\mathcal{Z}_t)$, defining a differentiated predictive model that allows the Bayes operator to reweight the posterior belief $\mu_{t+1}(\theta)$ after each realization of ζ_t. It is crucial to note that the participation (IR) and incentive compatibility (IC) constraints, in the sense of Myerson (1979) and Bolton and Dewatripont (2005), do not replace this process, but rather anchor the functional form of ℓ_t by restricting the space of admissible strategic profiles; only under a profile that satisfies these conditions does the observed likelihood coincide with that induced by an implementable mechanism, thereby guaranteeing that the informational update is numerically stable, structurally identified, and fully consistent with PBE.

The belief process is represented as a martingale under the ex-ante law. The following definition formalizes the consistency of this process with the PBE, a necessary condition for the Bayesian update to depend not on the specified model but on the structural experiment of the mechanism.

Definition 3 (Beliefs and PBE-consistent Bayesian updating)**.** A sequence of public beliefs about the captor's type must coincide with the Bayesian updating induced by the same PBE that generates the observed signals. Let $\mu_0 \in \Delta(\Theta_K)$ be the common prior over θ_K, with full conditional support on the initial public information: $\mu_0(\theta \mid z, \theta_V) > 0$ for all $\theta \in \Theta_K$. This prior summarizes the initial belief given the public location z and the victim's profile θ_V, so that

$$\mu_0(\theta \mid z, \theta_V) = \mathbb{P}_{\mathrm{E}}(\theta_K = \theta \mid z, \theta_V),$$

in accordance with the formalization of the structural prior in Section 2. Given a PBE, the strategic profile simultaneously determines the state's public policy, private strategies, action materialization through the MDG process, outcomes, and the ex-ante law $\mathbb{P}_{\mathrm{E}}$ over public trajectories. For $t \geq 1$, let $\mu_t \in \Delta(\Theta_K)$ denote the posterior obtained by iterated Bayes from μ_0, using the location z, the signals already observed $(\zeta_0, \ldots, \zeta_{t-1})$, and the predictive kernels $\ell_u(\theta'; J)$ induced by that same PBE. Therefore, updating with those realizations is equivalent to conditioning on the public filtration available at the opening of period t:

$$\mathcal{F}_t^{\text{pub}} = \sigma(z, \zeta_0, \ldots, \zeta_{t-1}),$$

The process $\{\mu_t\}_{t\geq 0}$ is said to be consistent with the PBE if, under the ex-ante law $\mathbb{P}_{\text{E}}$ induced by the same equilibrium,

$$\mu_t(\theta) = \mathbb{P}_{\text{E}}\big(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}}\big) \qquad \forall\, \theta \in \Theta_K,\ \forall\, t \geq 1, \tag{51}$$

where the equality is interpreted $\mathbb{P}_{\text{E}}$-almost surely upon fixing regular versions of the conditional probabilities. It is assumed that $\theta_K : (\Omega, \mathcal{F}) \to (\Theta_K, \mathcal{B}(\Theta_K))$ is measurable and that $\{\theta\} \in \mathcal{B}(\Theta_K)$ for all $\theta \in \Theta_K$; in particular, if Θ_K is finite, it suffices to take the power set σ-algebra. Thus, $\mathbf{1}\{\theta_K = \theta\}$ is measurable and integrable under $\mathbb{P}_{\text{E}}$, and the posterior μ_t coincides with the conditional marginal of the type given the observed public history. Furthermore, μ_0 coincides with the initial marginal of θ_K once conditioned on (z, θ_V), before the signals of the episode refine the structural prior.

□

Definition 3 implies that $\mu_t(\theta)$ is a bounded martingale under $\mathbb{P}_{\text{E}}$, meaning that the current belief is the unbiased prediction of the future belief given the accumulated public history. The following lemma formalizes the almost sure convergence that follows, relying on the convergence theorem of Doob (1953). Its proof is presented in Appendix B.

Lemma 4 (Asymptotic stability of beliefs)**.** *Under the ex-ante law $\mathbb{P}_{\text{E}}$, the belief process $\{\mu_t(\theta)\}_{t\geq 0}$ is a bounded martingale that converges $\mathbb{P}_{\text{E}}$-almost surely to a limit random variable Z_θ with $Z_\theta \in [0,1]$ $\mathbb{P}_{\text{E}}$-a.s.*

□

To establish the asymptotic properties of the learning process under the true probability measure $\mathbb{P}_{\theta_0}$, a fundamental tension must be resolved regarding the conditional independence of the observable history. In the live perfect Bayesian equilibrium, the planner's active dual control couples the policy instruments sequentially with past realized signals, as in Aghion et al. (1991) and Easley and Kiefer (1988). However, structural identification can be formally established by evaluating the system's filtering capacity over a frozen, deterministic policy path, following Blackwell and Dubins (1962). This conceptual device is formalized here under the notion of an informational slicing of the game tree, deferring the formal measure-theoretic proofs and regularities to Appendix B, where the mutual singularity of infinite product measures is derived via the product-measure criterion of Kakutani (1948).

Before defining the asymptotic event, it is useful to separate two planes of the model. Following the distinction between structure and measurement in dynamic estimation of Rust (1987) and Magnac and Thesmar (2002), the measures $\mathbb{P}_{\mathrm{C},\theta}$ used in the following definitions and in the Identification Theorem refer to the strategic competing-risks block, $j \in \{1, 2, 3\}$. In that domain, $\lambda_4 = 0$ is fixed, so the residual closure channel does not determine asymptotic type identification. In the data and in calibration, by contrast, $\lambda_4 > 0$ captures type-invariant residual closures; its contribution is therefore treated as an orthogonal component of the likelihood, in the sense of Lancaster (1990) and Abbring and Berg (2003). This separation allows the singularity criterion of Kakutani (1948) to operate on the signals generated by strategic risks. For the convergence of Lemma 4 to concentrate on the true type rather than a diffuse limit, the analysis is restricted to the event in which captivity generates an infinite sequence of informative public signals.

Definition 5 (Prolonged captivity event)**.** Let τ be the stopping time defined in (3). The prolonged captivity event is defined as

$$\mathcal{T} := \{\tau = \infty\} = \big\{\omega \in \Omega : m_t(\omega) \in \mathcal{M}_{\text{cont}} \ \forall\, t \geq 0\big\}.$$

In $\mathcal{T}$, the episode does not admit a terminal outcome in any finite period and generates an infinite sequence of public signals $\{\zeta_t\}_{t\geq 0}$.

□

Asymptotic identification requires that types other than the true type cannot sustain the same signal trajectory in $\mathcal{T}$, either because they cannot maintain captivity indefinitely or because their signal laws are mutually singular with those of the true type.

Definition 6 (Asymptotic identification in the event $\mathcal{T}$)**.** The model is asymptotically identifiable in $\mathcal{T}$ if, for all $\theta \in \Theta_K$ with $\theta \neq \theta_K^*$, at least one of the following conditions is satisfied.

(a) Exclusion by survival. $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) = 0$.

(b) Tail singularity of signals. $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) > 0$ and the laws of $\{\zeta_t\}_{t\geq 0}$ restricted to $\mathcal{T}$ under $\mathbb{P}_{\mathrm{C},\theta}$ and $\mathbb{P}_{\mathrm{C},\theta_K^*}$ are mutually singular.

□

The heterogeneity in the risk coefficients $\beta_{K,j}(\theta_K)$ is the structural channel that operationalizes the separation of Definition 6, since different criminal organizations generate distinct survival and detection signatures in total variation, preventing sustained observational equivalence and activating the singularity criterion of Kakutani (1948). The state's ability to distinguish types in the asymptotic limit does not depend on a specific functional form, but on a fundamental property of information theory, namely that any family of predictive models that guarantees recurrent separation in total variation between its signal laws will induce mutual singularity of the trajectory measures and the consequent concentration of Bayesian beliefs. The following lemma specializes this principle to the case of competing risks, demonstrating how the architecture of Cox (1972) operationalizes this separation through the heterogeneity in the coefficients $\beta_{K,j}$. Its proof is presented in Appendix B.

Lemma 7 (Structural identifiability and asymptotic learning)**.** Reference framework. Part (i) relies on Cox (1972) and Bradburn et al. (2003) because the daily record Y_t is interpreted as the output of a proportional competing risks system, where the heterogeneity in $\beta_{K,j}(\theta)$ alters intensities, hazards, and masses on the same support, allowing types to be contrasted in total variation period by period. Part (ii) resorts to Kakutani (1948) and Shiryaev (2019) because, under conditional independence of the daily experiments and recurrent separation in total variation, the product of affinities vanishes, and the trajectory laws become mutually singular. Finally, Schervish (1995) fixes the language of Bayesian identification under which this singularity translates into the exclusion of false hypotheses and the concentration of the posterior on the true type.

Antecedents and necessary conditions (domain of contrast). The lemma relies on the finite menu Θ_K, the daily record $Y_t = (m_t, d_t)$, the public filtration $\{\mathcal{F}_t^{\text{pub}}\}$, and the predictive kernel of the mechanism, already formalized in Section 2 and in Definitions 1 and 3. In this Lemma, $p_\theta\big(\cdot \mid \mathcal{F}_t^{\text{pub}}\big)$ denotes the mass function of Y_t at the opening of t under the counterfactual type θ, induced by the fixed PBE.

(N1) Finite menu of types. Without $|\Theta_K| < \infty$, the learning cannot be parameterized as a finite mixture, nor can counterfactuals be compared type-by-type on the simplex $\Delta(\Theta_K)$.

(N2) Well-defined daily record. Without the PBE inducing $p_\theta\big(\cdot \mid \mathcal{F}_t^{\text{pub}}\big)$, there is no daily experiment on which to measure total variation.

(N3) Coherence with the mechanism. The laws p_θ must correspond to the daily draw and to the intensities/hazards of Section 2; otherwise, equation (52) would cease to be the operational link with $\beta_{K,j}$.

(i) Let $\theta, \theta' \in \Theta_K$ be distinct. If in a period t the observable executed triple $\tilde{A}_t = (\tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S)$ ((4)), the optimal instruments (α_t^*, γ_t^*), and the other covariates entering $\lambda_j(t)$ coincide under the counterfactual of replacing only the captor's type with θ or θ', and furthermore there exists $j \in \{1,2,3\}$ such that $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$, then the intensities $\lambda_j(t)$ differ between θ and θ' by the factor $\exp\big(\beta_{K,j}(\theta) - \beta_{K,j}(\theta')\big) \neq 1$; consequently, the aggregate closure $q(t)$, the causal mix $\xi_j(t)$, or the marginal hazards $\bar{h}_j(t) = q(t)\xi_j(t)$ are altered, and, barring exact degeneracies that equate all masses of Y_t, the distribution of m_t changes in total variation. More precisely, there exists $\varepsilon = \varepsilon(\theta, \theta', t) > 0$ such that

$$\big\| p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}}) - p_{\theta'}(\cdot \mid \mathcal{F}_t^{\text{pub}}) \big\|_{\text{TV}} \geq \varepsilon, \tag{52}$$

where $\| \cdot \|_{\text{TV}}$ is the total variation distance between the two laws on Y_t.

(C1) Local contrapositive. If for all $j \in \{1,2,3\}$ it were the case that $\beta_{K,j}(\theta) = \beta_{K,j}(\theta')$ in the fixed counterfactual, the multiplicative channel $\exp(\beta_{K,j}(\theta) - \beta_{K,j}(\theta'))$ does not discriminate; part (i) does not invoke type differences in this margin.

(C2) Knife-edges. Even with $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$, if exact compensations equate all masses of Y_t, the total variation can vanish; these are parametric arrangements excluded from the operational statement.

(ii) Fitted with a public policy path $(\chi_u)_{u\geq 0}$ (the state's reduced form), suppose that, conditional on this path and on $\mathcal{T}$, the observations $\{Y_t\}_{t\geq 0}$ are independent under $\mathbb{P}_{\mathrm{C},\theta}$ for each θ (equivalently, each Y_t follows a law $p_\theta^{(t)}$ that depends on t only via the already fixed policy). For each pair $\theta \neq \theta'$, let $\mathcal{S}_{\theta,\theta'} \subseteq \mathbb{N}$ be the set of periods t in which (52) holds with $p_\theta^{(t)}$ and $p_{\theta'}^{(t)}$. If, in the event $\mathcal{T}$ (Definition 5), the set $\mathcal{S}_{\theta,\theta_K^*}$ is infinite $\mathbb{P}_{\mathrm{C},\theta_K^*}$-a.s. and there exists $\varepsilon_\theta > 0$ such that $\|p_\theta^{(t)} - p_{\theta_K^*}^{(t)}\|_{\mathrm{TV}} \geq \varepsilon_\theta$ for all $t \in \mathcal{S}_{\theta,\theta_K^*}$, then the laws of $(Y_t)_{t\geq 0}$ conditioned on $\mathcal{T}$ and on the policy under $\mathbb{P}_{\mathrm{C},\theta}$ and under $\mathbb{P}_{\mathrm{C},\theta_K^*}$ are mutually singular.

(C3) Infiniteness and margin. Without an infinite subset of periods with $\mathrm{TV} \geq \varepsilon_\theta > 0$, the product of Hellinger affinities does not vanish, and the criterion of Kakutani (1948) does not, by itself, force singularity between the product laws.

(C4) Independence conditional on policy. This is a transparent regime sufficiency; outside of it, the law of (Y_t) may not factor as $\bigotimes_t p^{(t)}$, and an alternative route is required (e.g., recurrence of public states where (i) is reactivated with the same bound).

□

Combining the bounded martingale of Lemma 4, the asymptotic identification of Definition 6, and the structural separation of Lemma 7, the following theorem establishes that beliefs converge almost surely to the true type. Its proof is in Appendix B.

Theorem 8 (Identification and Consistency of True Types)**.** Consider a prior $\mu_0 \in \Delta(\Theta_K)$ with full support on the type space and a PBE-consistent Bayesian updating according to Definition 3. Let $\theta^* := \theta^*_K$ be the realized type and $\mu_C := \mathbb{P}_{\mathrm{C},\theta^*}(\cdot \mid \mathcal{T})$ the law of the true type restricted to the prolonged captivity event $\mathcal{T}$ (Definition 5) and renormalized.

If the mechanism is asymptotically identifiable in the event $\mathcal{T}$ under Definition 6, and since the bounded martingale structure of beliefs guarantees convergence to a random limit by Lemma 4, the following properties are satisfied μ_C-almost surely:

1. Exclusion of false hypotheses. The posterior mass assigned to any false type vanishes asymptotically.

$$\mu_t(\theta) \xrightarrow{t\to\infty} 0, \qquad \forall\, \theta \in \Theta_K,\ \theta \neq \theta^*.$$

   This property is met under the asymptotic sufficiency of Lemma 7, part (ii), where the recurrent separation in total variation and the singularity criterion of Kakutani (1948) ensure that the trajectory laws of different types are mutually singular Shiryaev (2019).

2. Concentration on the true type. The posterior mass of the realized type converges to unity.

$$\mu_t(\theta^*) \xrightarrow{t\to\infty} 1.$$

   This collapse is a direct consequence of the normalization of the probability measure on the finite simplex $\Delta(\Theta_K)$ after the exclusion of false types.

Consequently, on the event $\mathcal{T}$, the sequence of beliefs $\{\mu_t\}$ converges weakly to the Dirac measure δ_{θ^*}.

□

3. Mechanism implementability

3.1. Implementability

The implementability of the mechanism connects two distinct levels. The first is strategic consistency, understood as the existence of a profile in which the state, the captor, and the family are sequentially rational and update beliefs via Bayes' rule at all reached nodes. The second is the sustainability of the public outcome, that is, that the state's social choice rule, along with the instruments (α_t, γ_t), simultaneously satisfies IC^K, IR^K, and IR^F. Therefore, the following definitions and Theorem 11 are crucial, as they guarantee that the policy subsequently calibrated is not only informative for type identification but also executable as an equilibrium path when the captor's type is private and the family's participation cannot be mechanically imposed.

Definition 9 (Assessment and PBE)**.** Consider a discrete-time Bayesian game with horizon $\mathbb{T} = \{0, 1, \ldots, \tau\}$, players $\mathcal{J} = \{S, K, F\}$, and captor's type $\theta_K \in \Theta_K$ according to a common prior $\mu_0 \in \Delta(\Theta_K)$ with $\mu_0(\theta) > 0$ for all $\theta \in \Theta_K$. Following Fudenberg and Tirole (1991a) and Kreps and Wilson (1982), an assessment in this dynamic game is described by the following formal objects:

- *Strategy profile*, which is given by $s = (s^S, s^K, s^F)$, where $s^S = \chi = (\chi_t)_{t\in\mathbb{T}}$ with

  $$\chi_t \colon H_t \times \mathcal{R} \longrightarrow \mathcal{A}^{S,\text{disc}} \times \mathcal{A}^{\alpha} \times \mathcal{A}^{\gamma}, \qquad \mathcal{A}^{S,\text{disc}} = \{\text{Rescue, Negotiate}\}.$$

  The profiles s^K and s^F are behavioral strategies of the private agents in their respective information sets.

- *Belief system*, $\mu = \{\mu_t\}_{t \geq 0}$, with $\mu_t \in \Delta(\Theta_K)$, summarizing the uncertainty regarding θ_K after what has been publicly observed up to the opening of t.

- *Ex-ante law*, $\mathbb{P}^s_{\mathrm{E}}$, which defines the probability law of the episode induced by the profile s, the prior μ_0, and the physical dynamics of captivity.

Perfect Bayesian Equilibrium. The pair (s, μ) is a PBE if conditions (a) and (b) hold.

(a) Sequential rationality. In every t and in every information set reached with strictly positive probability under $\mathbb{P}^s_{\mathrm{E}}$, the equilibrium actions of each agent $i \in \mathcal{J}$ solve the optimization program conditioned on $\mathcal{I}^i_t$, given μ_t and s^{-i}. For the state, the optimization is:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t)} \Big\{ \sum_{\theta \in \Theta_K} \mu_t(\theta)\, L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi^S_{t, a^S}(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\},$$

with the selection rule $a_t^{S*} =$ Rescue if the expected cost of rescue does not exceed that of negotiation, and $a_t^{S*} =$ Negotiate otherwise, consistent with the discrete rule (46). For each private agent $i \in \{K, F\}$:

$$a_t^{i*} \;\in\; \arg\max_{a_t^i \in \mathcal{A}^i} \mathcal{U}_t^i\big(a_t^i \mid \mathcal{I}_t^i, \chi, s^{-i}\big),$$

where L_t is the type-conditional social loss, Π^S_{t,a^S} is the penalty for deviation from the Bayesian benchmark of the chosen branch, $\Delta H_t(\alpha_t, \gamma_t)$ is the expected information gain, and $\mathcal{U}_t^K, \mathcal{U}_t^F$ are the expected utilities of the captor and the family in t.

(b) On-path Bayesian consistency. In every node reached with positive probability under $\mathbb{P}^s_{\mathrm{E}}$, beliefs coincide with the Bayesian inference from μ_0 and the observed public signals. For all $\theta \in \Theta_K$ and all $t \geq 1$,

$$\mu_t(\theta) = \mathbb{P}^s_{\mathrm{E}}\big(\theta_K = \theta \mid \mathcal{F}^{\mathrm{pub}}_t\big), \qquad \mathcal{F}^{\mathrm{pub}}_t = \sigma(z, \zeta_0, \ldots, \zeta_{t-1}), \tag{53}$$

where $\mathbb{P}^s_{\mathrm{E}}$ is the ex-ante law induced by the strategy profile s, the common prior μ_0, and the physical dynamics of captivity. The definition is read as a fixed-point condition, whereby given s, the law $\mathbb{P}^s_{\mathrm{E}}$ determines the beliefs by Bayes at reached nodes; given this belief system, s must be sequentially optimal. At $t = 0$, μ_0 is the common prior conditioned on the observable geographic zone. Off the equilibrium path, beliefs are admissible and compatible with sequential rationality in the remaining subgames Fudenberg and Tirole (1991a); Kreps and Wilson (1982); the subsequent result is one of existence, not uniqueness, and therefore admits the standard off-path multiplicity in dynamic games with incomplete information.

□

Definition 10 (Nash-Bayes-IC^K,IR^K,IR^F Implementability)**.** Let (s^*, μ^*) be a PBE in the sense of Definition 9. The mechanism is implementable Nash-Bayes-IC^K,IR^K,IR^F if the equilibrium profile additionally satisfies the following three feasibility conditions.

Bayesian incentive compatibility of the captor (IC^K). For all $t \in \mathbb{T}$ and all rival mimicry $\theta_j \in \Theta_K$,

$$\mathbb{E}_{\theta \sim \mu^*_t}\left[V^K_t\big(a^*(\theta) \mid \theta, \alpha^*_t, \gamma^*_t\big) - V^K_t\big(a^*(\theta_j) \mid \theta, \alpha^*_t, \gamma^*_t\big)\right] \geq 0,$$

where $a^*(\theta)$ denotes the trajectory of equilibrium actions of type θ under policy χ^*. This is the global non-mimicry condition in expected value, where the expectation is taken over the true type, and the deviation θ_j remains fixed.

Bayesian participation constraint of the captor (IR^K). This constraint is not a reservation rent in the sense of Myerson (1979); in kidnapping for ransom, the interaction is coercive and not contractual. It requires that the utility of release dominates in expected value the best criminal alternative between continuation and killing, evaluated at the equilibrium instruments:

$$\mathbb{E}_{\theta\sim\mu_t^*}\left[U^K_{\text{rel}}(\theta)-\max\left\{V^K_{\text{cont},t}(\theta,\alpha_t^*,\gamma_t^*),U^K_{\text{kill}}(\theta,\theta_S)\right\}\right]\geq 0.$$

The reading is consistent with the logic of Becker (1968) adapted to armed conflict, whereby public policy erodes the expected profitability of persisting or escalating violence relative to surrender. Here $V^K_{\text{cont},t}$ is the continuation component of the value function V_t^K used in IC^K, evaluated on the branch where captivity has not been absorbed, according to (34).

Participation constraint of the family (IR^F).

$$\mathcal{U}_t^F(a_{\text{coop}})\ \geq\ \mathcal{U}_t^F(a_{\text{col}}),$$

evaluated with the information $\mathcal{I}_t^F$ and the reference equilibrium (s^*,μ^*), according to (49). The simultaneous compatibility of this constraint with IC^K and IR^K is not assumed separately. The set of policy triples that simultaneously satisfy the three preceding conditions is called the feasible set of policies and is denoted by

$$\Gamma_t(\mu_t):=\left\{(a_t^S,\alpha_t,\gamma_t)\in\mathcal{A}^{S,\text{disc}}\times[0,1]\times[0,1]:IC^K,\ IR^K,\ IR^F\text{ are satisfied}\right\}. \tag{54}$$

Building on the existence framework of Kadan et al. (2017), Theorem 11 treats the non-emptiness of $\Gamma_t(\mu_t)$ as the primitive robust-feasibility condition ensuring the existence of an implementable PBE. The proof is presented in Appendix B.

□

Theorem 11 (Conditional implementability of the mechanism in PBE)**.** Consider the PBE of Definition 9 under the following conditions:

(i) $|\Theta_K| < \infty$, with strictly positive prior $\mu_0(\theta) > 0$ for all $\theta \in \Theta_K$.

(ii) The action spaces $\mathcal{A}^i$ are finite for all $i \in \{K, S, F\}$ and the game tree has a finite number of nodes in each period t, so that the behavioral strategy space of each agent is a finite product of simplexes.

(iii) The implementation law $\mathbb{P}_{\mathrm{I},i}$ of the MDG process assigns strictly positive probability to every action $a \in \mathcal{A}^i$ for every plan a_t^{i*} and state X_t (full support, (4)), which corresponds to the base specification with $\underline{c} > 0$.

(iv) The instrument sets are $\mathcal{A}^\alpha = [0,1]$ and $\mathcal{A}^\gamma = [0,1]$, compact and convex; the functions L_t and $\mathcal{U}_t^i$ are measurable, uniformly bounded in $\mathbb{T}$, and continuous in the continuous decision variables. The physical outcome rule G_t is measurable, and the transition kernel Q_t induced by G_t and by the competing-risk probabilities is continuous in those variables.

(v) Robust feasibility. For each t and each $\mu_t \in \Delta(\Theta_K)$, the feasible set $\Gamma_t(\mu_t)$ in (54) is nonempty and compact. For each discrete branch a_t^S, its continuous section

$$\Gamma_t^{a^S}(\mu_t) := \{(\alpha_t, \gamma_t) : (a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t)\}$$

is convex, and the correspondence $\mu_t \mapsto \Gamma_t(\mu_t)$ is continuous (upper and lower hemicontinuous) and compact-valued in $\Delta(\Theta_K)$. Non-emptiness is a primitive condition of simultaneous compatibility among IC^K, IR^K, and IR^F, according to Kadan et al. (2017).

(vi) Parametric separability. For all $t \in \mathbb{T}$, the captor's value function admits the representation

$$V_t^K\big(a^*(\theta') \mid \theta_K, \alpha_t, \gamma_t\big) = \sum_j w_j\big(a^*(\theta'), \alpha_t, \gamma_t\big) \cdot \exp\big(\beta_{K,j}(\theta_K) \cdot z\big),$$

where the weights $w_j(\cdot)$ depend on the action path $a^*(\theta')$ and the instruments, but not on θ_K. This condition allows auditing the economic direction of the incentives; it does not replace the verification of IC^K in (v) nor the statistical identification of Lemma 7.

(vii) Quasiconvexity of the restricted objective. For each t, each $\mu_t \in \Delta(\Theta_K)$, and each discrete branch $a_t^S \in \mathcal{A}^{S,\text{disc}}$, the functional

$$\begin{aligned}(\alpha_t, \gamma_t) \ \mapsto \ &\sum_{\theta_K \in \Theta_K} \mu_t(\theta_K)\, L_t(a_t^S, \alpha_t, \gamma_t, \theta_K, \iota_t) \\ &+ \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t)\end{aligned}$$

is quasiconvex in (α_t, γ_t) over $\Gamma_t^{a^S}(\mu_t)$. Combined with the convexity of $\Gamma_t^{a^S}(\mu_t)$ (hypothesis (v)), this condition guarantees that the optimal restricted correspondence $\Psi_t(\mu_t) := \arg\min_{\Gamma_t^{a^S}(\mu_t)}\{\cdot\}$ is convex-valued, which makes Kakutani's fixed-point theorem applicable to the state's program. It is satisfied in particular when the complete objective, including the informational motive $-\psi_H \Delta H_t$, preserves quasiconvexity in (α_t, γ_t); for example, when L_t and Π_{t,a^S}^S are convex, as is the case with the type-specific quadratic functionals C_{ops} and C_{maint} of the model and with the quadratic deviation penalty, and the entropy term does not destroy the geometry of the relevant sublevel sets.

Then there exists a pair (s^*, μ^*), with $s^* = (\chi^*, s^{K*}, s^{F*})$, that satisfies the following three conditions.

(a) Constitutes a PBE in the sense of Definition 9: the strategies in s^* are mutual best responses given μ^*, and the beliefs μ_t^* are consistent with Bayes' rule at reached nodes with positive probability.

(b) Implements the social choice rule f_t: on the equilibrium path, $\chi_t^*(h_t) = (a_t^{S*}, \alpha_t^*, \gamma_t^*)$ coincides with the allocation that minimizes the expected social loss (45) subject to $\Gamma_t(\mu_t^*)$, in accordance with rule (46).

(c) Satisfies IC^K, IR^K, and IR^F under the profile (s^*, μ^*).

□

Theorem 11 bridges learning and strategy by complementing identification. The Kakutani route establishes that sustained pressure can make daily signals discriminate types; this theorem supplies the strategic counterpart, showing that the state can deploy such pressure on an equilibrium path without violating IC^K, IR^K, or IR^F, so the captor does not rationally exit through premature lethal closure and the family does not abandon the mechanism. The guarantee is conditional on robust feasibility, hypothesis (v), verified in the calibrated scenarios though not assured a priori along every belief trajectory. While $\Gamma_t(\mu_t)$ remains non-empty and negotiation incentives dominate the lethal outside option, an implementable PBE keeps adversary conduct predictable enough for evidence on θ_K^* to accumulate. The calibration that follows therefore evaluates a trajectory that is simultaneously strategic and informative.

3.2. *Economic policy implications*

The implications for economic policy concentrate on how the state should adjust operational pressure when learning about the captor's type. The pressure γ_t does not have a mechanical, uniform effect on the duration of captivity, since its impact depends on whether it accelerates the closing channels more than it shifts or inhibits other outcomes. Furthermore, when the posterior assigns greater mass to high-risk types, optimal policy must respond more intensely only if the net marginal effect of the full program favors greater pressure. The following two propositions formalize these margins, where the first links Bayesian identification and optimal pressure as a local result of the state's complete objective, and the second characterizes how pressure alters the expected duration of captivity.

The first proposition converts Bayesian learning into a local operational rule, whereby if the public history shifts the posterior toward a type with greater propensity for critical outcomes, state pressure increases when material loss, the deviation premium, and informational value jointly point in that direction. The dynamic acquisition of information of Zhong (2022) provides the foundation for treating information as an intertemporal choice object, since the decision maker may accept current learning costs when present instruments modify the quality of future signals and, therefore, the value of later decisions. In this model, that intuition is incorporated through $-\psi_H \Delta H_t$ and the net monotonicity condition (56).

Proposition 12 (Optimal operational pressure and Bayesian type identification)**.** *Suppose $\Theta_K = \{\theta_{low}, \theta_{high}\}$ and fix a discrete branch $b \in \{R, N\}$ in a neighborhood of the posterior belief considered. Let $x_t = (\alpha_t, \gamma_t)$ and define the complete objective of the branch*

$$J_t^b(x_t, \mu_t) := \sum_{\theta \in \Theta_K} \mu_t(\theta) L_t(b, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,b}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t). \tag{55}$$

If (α_t^*, γ_t^*) is an interior optimum of $J_t^b(\cdot, \mu_t)$, the constraints IC^K, IR^K, and IR^F do not locally change regime, the Hessian $H_t^b := \nabla_{xx}^2 J_t^b(x_t^*, \mu_t)$ is positive defi-

nite, and the net marginal effect of increasing $\mu_t(\theta_{high})$ on the first-order condition for γ_t favors greater pressure, that is,

$$e'_\gamma \big(H_t^b\big)^{-1} \nabla^2_{x\mu} J_t^b(x_t^*, \mu_t) < 0, \qquad e_\gamma = (0,1)', \tag{56}$$

then the optimal operational pressure of that branch is locally increasing in the posterior belief over the high-risk type:

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} > 0.$$

The structural separation $\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})$ for $j \in \{2,3\}$ contributes to condition (56) through the ratio

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp\!\big(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})\big), \quad j \in \{2,3\},$$

but the monotonicity of γ_t^* depends on the joint net effect of material losses, deviation penalty, and informational gain.

□

The second proposition translates that pressure rule into a directly interpretable magnitude for public policy, namely the expected duration of captivity. Its importance is that it separates the cases in which intensifying the operation accelerates closure from those in which it may prolong negotiation by displacing the payment channel without sufficiently compensating for the risks of rescue or death. In this sense, the proposition provides the physical translation of the dynamic information design of Zhong (2022). While there the optimal acquisition of information depends on the frequency, precision, and temporal cost of signals, here the net sensitivity κ_h shows how operational pressure alters the probabilistic frequency of outcomes. State intervention can accelerate type revelation by modifying closing hazards, but may also accept the risk of prolonging captivity when that temporal cost allows the structural fingerprint of the captor to be more clearly isolated.

Proposition 13 (Expected duration of captivity and operational pressure)**.** *Standing assumptions. Two working assumptions are imposed.* (S1) *The derivative $\partial\mathbb{E}[\tau \mid \theta_K]/\partial\gamma_t^*$ is a partial derivative: γ_t^* is perturbed in a single period t while the instrument sequence $\{\gamma_u^*\}_{u\neq t}$ is held fixed; expected duration is expressed as $\mathbb{E}[\tau \mid \theta_K] = \sum_{u=0}^{\infty} S(u \mid \theta_K)$ with $S(u \mid \theta_K) = \prod_{r=1}^{u} p_{\text{Cont},r|\theta_K}$, so only factors $S(u)$ with $u \geq t$ depend on γ_t^*.* (S2) *Operational pressure shifts each intensity proportionally to its level: $\partial\tilde{\lambda}_j(t \mid \theta_K)/\partial\gamma_t^* = \zeta_{\gamma,j}\,\tilde{\lambda}_j(t \mid \theta_K)$, where $\zeta_{\gamma,j} := \partial \ln \tilde{\lambda}_j(t \mid \theta_K)/\partial\gamma_t^*$ is the semi-elasticity of intensity j with respect to γ_t^*; λ_4 is invariant to γ_t^* because it is exogenous.*

Define the net sensitivity of type θ_K to the optimal operational pressure γ_t^* as follows.

$$\kappa_h(\theta_K, t) := \Big(\zeta_{\gamma,2}\,\tilde{\lambda}_2(t \mid \theta_K) + \zeta_{\gamma,3}\,\tilde{\lambda}_3(t \mid \theta_K)\Big) - \zeta_{\gamma,1}\,\tilde{\lambda}_1(t \mid \theta_K). \tag{57}$$

If $\kappa_h(\theta_K, t) > 0$ for all t, the expected duration of captivity $\mathbb{E}[\tau \mid \theta_K]$ is strictly decreasing in γ_t^*. If $\kappa_h(\theta_K, t) < 0$ for all t, it is strictly increasing. The model thus generates the following testable differential prediction:

$$\operatorname{sgn}\left(\frac{\partial\mathbb{E}[\tau \mid \theta_K]}{\partial\gamma_t^*}\right) = -\operatorname{sgn}(\kappa_h(\theta_K, t)), \tag{58}$$

with the sign varying across types through the structural parameters $\beta_{K,j}(\theta_K)$.

□

4. Application: Kidnapping for Ransom in Colombia

Having defined the mechanism and its implementation, this section draws on microdata to calibrate the prior beliefs (μ_0) and to establish the correspondence between the data and the model's primitives. More than a case study, this section shows that the game's structural objects, priors, hazards, instruments, and the feasible policy set, admit rigorous identification from observed microeconomic variation.

4.1. Empirical anchoring of the parameter space

Between 1964 and 2025, Colombia registered 61,341 kidnapping cases, according to CNMH (2026); although large parts of the country suffered this crime during this period, cases concentrated in a set of municipalities with high urban centrality or strategic value, led by Bogotá (1,437 cases), Cali (784), and Medellín (652), followed by Valledupar, Argelia, Villavicencio, Aguachica, Ocaña, Bucaramanga, and Santa Marta, tied to successive phases of the internal armed conflict, from the early consolidation of insurgent groups through the 1996 creation of the Unified Action Groups for Personal Liberty, GAULA, established under CRC, Law 282 (1996), to a marked post-2011 hardening of institutional deterrence documented by CNMH (2013). This concentrated, spatially clustered pattern is consistent with the crime triangle framework of Pires et al. (2014), which documents spatio-temporal concentrations of kidnapping for ransom in Colombia across its three vertices: victims, offenders, and places. The historical peak in the late 1990s and early 2000s coincided with the expansion of the armed conflict and illegal economies, as shown by Pinto et al. (2004), while Zárate (2007) documents kidnapping's use as a financing and political-coercion mechanism. Both elements motivate treating criminal heterogeneity as a structural feature of the phenomenon rather than an estimation nuisance. Figures I and II illustrate, respectively, this temporal and municipal concentration and the Colombian trajectory across administrations, conflict phases, and legal milestones.

(a) Temporal evolution of kidnappings

(b) Municipal geographic distribution

Figure I: Temporal evolution and geographic distribution of kidnappings in Colombia.

Note: Panel (a) shows the evolution of the total number of kidnapping cases in Colombia, 1964–2025, highlighting the main phases of escalation and decline. Panel (b) presents the geographic distribution of kidnapping cases by Colombian municipalities, 1964–2025. *Source:* Author's calculations based on CNMH (2026).

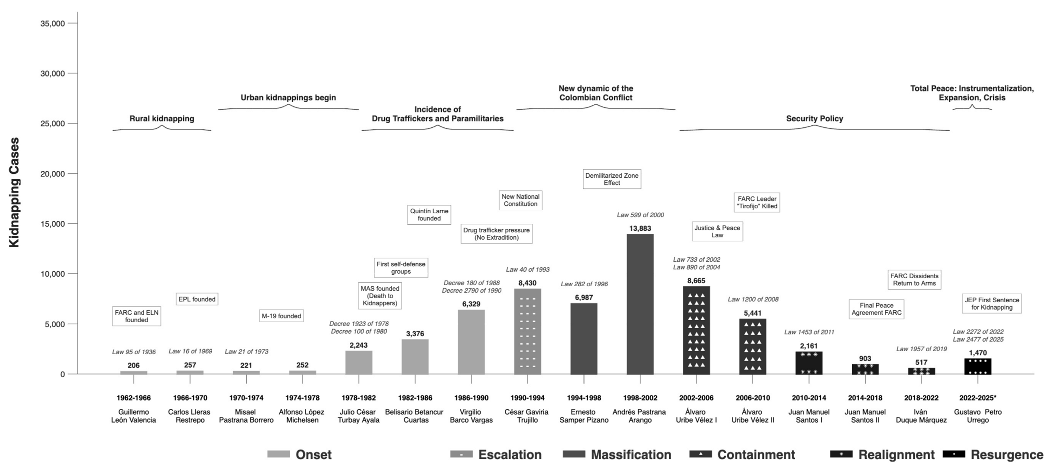
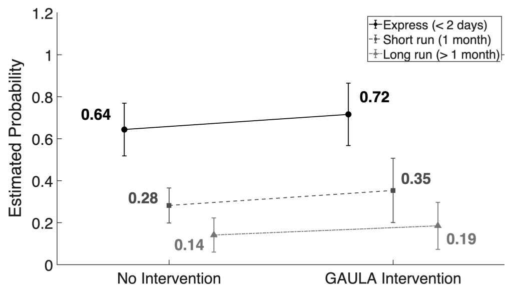


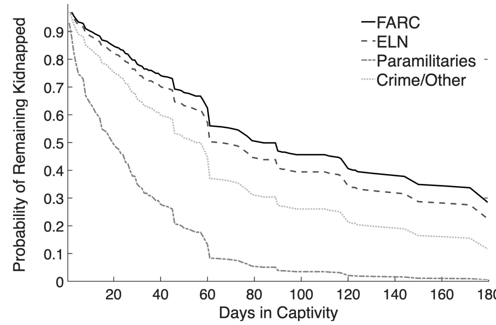
Figure II: Evolution of kidnappings in Colombia: armed conflict, security policy, and institutional change (1962–2025).

Note: The figure presents the total number of kidnappings per presidential term in Colombia from 1962 to 2025. Shading distinguishes phases of the armed conflict, and annotations highlight developments related to armed groups, legal reforms, and security policies. *Source:* Author's elaboration based on CNMH (2026) and Pinto et al. (2004).

Through a Multinomial Logit (MNL) model and a cause-specific proportional hazards model of Cox (1972), this organizational-level heterogeneity is quantified. Both models are estimated on $n = 1{,}125$ individual-level cases from the National Center for Historical Memory, according to CNMH (2026), with full specification, robustness checks, and sample-selection diagnostics as noted in Appendix A. The MNL's static marginal probability of mortality in captivity ranges from 0.002 for the FARC to 0.071 for common delinquency (DC). Payment hazard ratios relative to the FARC reach 4.28 for paramilitaries (PAR) and 1.71 for DC, while the ELN is statistically indistinguishable from the reference group. The cumulative probability of tactical rescue by day 100 varies between 0.27 and 0.33 by type; after 2011, the rescue hazard multiplies by 4.86 and the payment hazard falls by 49%, consistent with the strengthening of the GAULA's deterrent capacity, as shown in Figure III. The dynamic mortality profile from Cox in Figure IV reproduces the same ordering across organizations, with comparable magnitudes (0.002 to 0.079). These estimates supply the mechanism's type menu $\Theta_K = \{\text{DC}, \text{PAR}, \text{ELN}, \text{FARC}\}$, order the reservation utility of release $U^K_{\text{rel}}(\theta)$ across types, and anchor the type-specific effective intensities $\tilde{\lambda}_j(t \mid \theta_K)$ whose empirical separation is the observational counterpart of the total-variation separation hypothesis in Lemma 7.



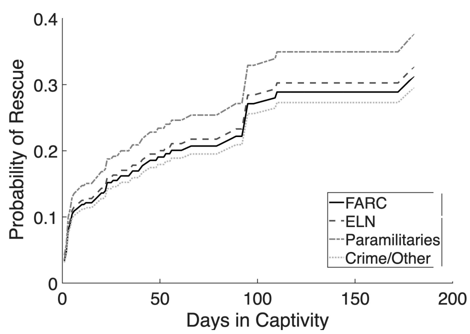
(a) Rescue by GAULA intervention and duration



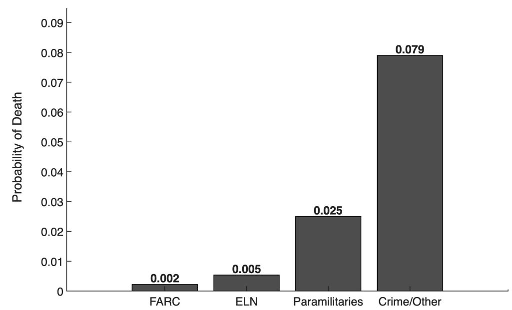
(b) Survival of duration by perpetrator

Figure III: Temporal dynamics of rescue and heterogeneity in captivity duration.

Note: Panel (a) reports the probability of Rescue by GAULA intervention and duration category. Panel (b) presents the survival curves of captivity duration by perpetrator group. *Source:* Author's calculations based on CNMH (2026).



(a) Cumulative state Rescue probability



(b) Mortality in captivity by perpetrator group

Figure IV: Cumulative rescue probability and mortality in captivity by armed group.

Note: Panel (a) shows the cumulative state Rescue probability by armed group, 1964–2025. Panel (b) reports the probability of Death in captivity by perpetrator group. *Source:* Author's calculations based on CNMH (2026).

4.2. *From estimators to structural inputs*

The MNL and the Cox model connect microeconomic evidence with the initial distribution $\mu_0 \in \Delta(\Theta_K)$ and the technology vectors $\beta_{K,j}(\theta_K)$. The MNL provides the static fingerprint of each type by relating perpetrator, victim, context, and tactics to the conditional distribution of outcomes; thus, the initial belief and terminal payments are set with observed weights. The Cox model provides the dynamic dimension: its cause-specific hazards are transferred to the mechanism as rates $\lambda_j(\theta_K, t)$ that govern the daily signal ζ_t, update the posterior μ_t, and determine the speed of learning. The empirical separation of those rates across organizations is the observational basis of Lemma 7 and the criterion of Kakutani (1948). Whenever the trajectories induced by the types generate distinct laws, the posterior can concentrate on the true type. Together, this microeconometric block converts case-level data into priors, costs, hazards, and constraints directly usable by the dynamic mechanism. The explicit translation protocol between estimators and structural parameters is detailed at the beginning of Section 5.

5. Calibration of the mechanism and simulations

This section calibrates the specialized mechanism of Section 4 in four simulated episodes of finite horizon, holding fixed the general architecture of Section 2. The exercise combines microeconometric priors and diagnostics to study four objects, which comprise the evolution of beliefs $\mu_t(\theta)$, the signals observed by the state, the response of the instruments (α_t, γ_t), and compliance with the IR and IC constraints. Establishing these four objects requires solving the entire mechanism through backward induction, which is why the model turns to artificial intelligence, as solving the Bellman equation for the kidnapper's continuation branch through traditional optimization is computationally costly, since it requires recursively recalculating the expected value of sustaining captivity in every period; this same recursive cost propagates to the solution of the entire mechanism, since the state's dual-control problem needs that continuation value at every period to determine the optimal instruments that guarantee $\Gamma_t^{a^S}(\mu_t)$. To drastically reduce this computational cost and anticipate future decisions, the model employs a value approximator based on a multilayer perceptron (MLP), trained in two independent instances, where one approximates the continuation value that feeds the state's optimization, and the other approximates the kidnapper's optimal response under its true type. This neural-network approximation functions as an advanced simulator that analyzes 34 environmental variables to predict how much value each actor assigns to continuing the interaction. The network was trained using deep-learning algorithms that evaluated 2,000 computer-generated possible scenarios. In practice, the system performs this calculation in real time to optimally determine both the captor's response and the state's policy under different levels of pressure and control instruments. The complete mathematical details of the architecture, the network configuration, and the computational calibration process are developed in full in Appendix C.

To evaluate the model's dynamic performance, the calibration is initialized by selecting a set of initial parameters and covariates that define the kidnapping environment. In this design, the modeler exogenously determines the captor's private type, which constitutes the ground-truth benchmark for measuring learning. Nonetheless, consistent with the assumption of incomplete information, the dynamic mechanism operates such that both the state and the family are unaware of that private type, interacting and choosing their optimal strategies solely on the basis of their beliefs or Bayesian priors.

Scenario I corresponds to DC, with $\theta_K^* = \mathrm{DC}$, lax state with pressure on crime, high-wealth family, private victim, and Metropolis location; *Scenario II* corresponds to PAR, with $\theta_K^* = \mathrm{PAR}$, strict state, standard-wealth family, public victim, and Caribbean location; *Scenario III* corresponds to ELN, with $\theta_K^* = \mathrm{ELN}$, strict state, high-wealth family, private victim, and Eastern Plains/Jungle location; and *Scenario IV* corresponds to FARC, with $\theta_K^* = \mathrm{FARC}$, lax state with pressure on crime, high-wealth family, public victim, and Pacific / Red Zone location. All four are simulated over a fixed horizon $\bar{T} = 150$ periods (days), and optimal policy is computed from the state's minimization of social cost. In the order $(\mathrm{DC}, \mathrm{PAR}, \mathrm{ELN}, \mathrm{FARC})$, the initial priors are $\mu_0^{\mathrm{DC}} = (0.35, 0.35, 0.15, 0.15)$, $\mu_0^{\mathrm{PAR}} = (0.15, 0.15, 0.25, 0.45)$, $\mu_0^{\mathrm{ELN}} = (0.05, 0.35, 0.25, 0.35)$, and $\mu_0^{\mathrm{FARC}} = (0.35, 0.35, 0.15, 0.15)$. Tables I and II present the structured summary of the four scenarios evaluated, detailing, respectively, the configuration of the contextual covariates of state, family, victim, and location, as well as the initial beliefs (*priors*) assigned to each captor type.

The posterior of the true type allows evaluating the mechanism's capacity to separate types from the sequence of signals. As reported in Figure V, posterior mass converges gradually toward the true type in all four scenarios, reaching $\mu_{150}(\theta_K^*) \approx 1{,}000$; this implies that the mechanism accumulates identification under different levels of wealth, regions, and visibility. The trajectories reflect dynamic Bayesian learning from the signals observed in each cycle, until belief mass definitively concentrates on the true perpetrator. The gray bars mark the realized stopping time; the subsequent counterfactual extension shows that the speed of learning and the moment of absorption are distinct objects of the model.

Table I: Parameters and covariates by calibration scenario

Scenario	State	Family	Victim	Location
I: DC	Lax	High wealth	Private	Metropolis
II: PAR	Strict	Standard	Public	Caribbean
III: ELN	Strict	High wealth	Private	Eastern Plains/Jungle
IV: FARC	Lax	High wealth	Public	Pacific / Red Zone

Source: Author's elaboration.

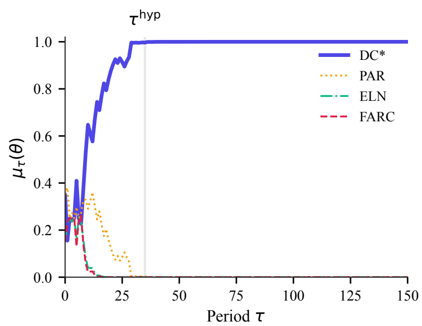
Table II: Initial beliefs (*priors*) by calibration scenario

Scenario	$\mu_0(\mathrm{DC})$	$\mu_0(\mathrm{PAR})$	$\mu_0(\mathrm{ELN})$	$\mu_0(\mathrm{FARC})$
I: DC	0.35	0.35	0.15	0.15
II: PAR	0.15	0.15	0.25	0.45
III: ELN	0.05	0.35	0.25	0.35
IV: FARC	0.35	0.35	0.15	0.15

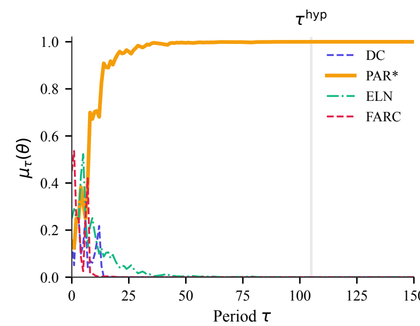
Source: Author's elaboration.

Taken together, the figure illustrates finite-horizon behavior, whereby posterior mass shifts toward the true type as signals accumulate separation, in the direction predicted by Theorem 8 under $\mathcal{T}$, without constituting direct evidence for the asymptotic result; learning consolidates under the institutional, territorial, and visibility environment of each case.

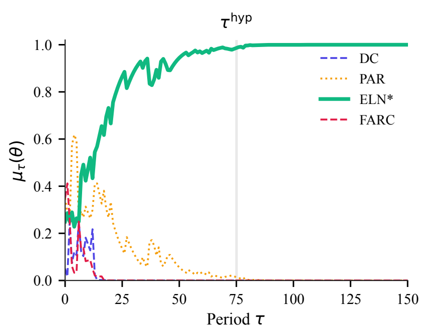
The optimal operational pressure allows evaluating how the State adjusts the intensity of intervention as it learns about the captor's type. Figure VI compares the trajectory of γ_t^* with the belief-weighted benchmarks of the Rescue (γ_R^μ) and Negotiation (γ_N^μ) branches. In the baseline, the rescue benchmark values by type



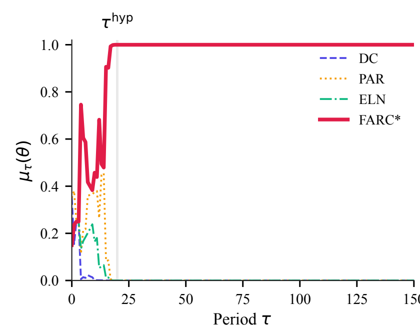
(a) Scenario I: DC



(b) Scenario II: PAR



(c) Scenario III: ELN



(d) Scenario IV: FARC

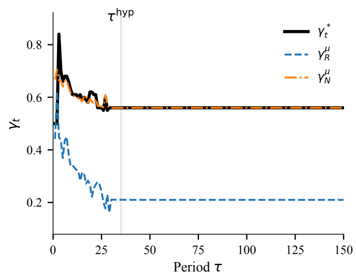
Figure V: Bayesian beliefs of each type $\mu_\tau(\theta)$ across scenarios.

Note: Panel (a) corresponds to Scenario I: $\theta_K^* = \text{DC}$, lax θ_S, high-wealth θ_F, private θ_V, and Metropolis z. Panel (b) corresponds to Scenario II: $\theta_K^* = \text{PAR}$, strict θ_S, standard-wealth θ_F, public θ_V, and Caribbean z. Panel (c) corresponds to Scenario III: $\theta_K^* = \text{ELN}$, strict θ_S, high-wealth θ_F, private θ_V, and Eastern Plains/Jungle z. Panel (d) corresponds to Scenario IV: $\theta_K^* = \text{FARC}$, lax θ_S, high-wealth θ_F, public θ_V, and Pacific / Red Zone z. The gray band marks τ^{hyp}, the first period in which the mechanism outcome differs from continuation. *Source:* Author's elaboration.

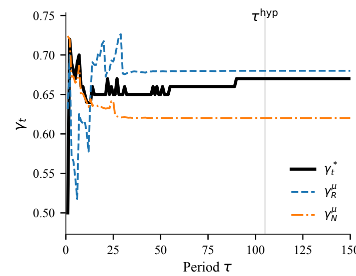
are $\gamma_R^{\text{DC}} = 0.23$, $\gamma_R^{\text{PAR}} = 0.37$, $\gamma_R^{\text{ELN}} = 0.50$, and $\gamma_R^{\text{FARC}} = 0.64$, while those for negotiation are $\gamma_N^{\text{DC}} = 0.59$, $\gamma_N^{\text{PAR}} = 0.64$, $\gamma_N^{\text{ELN}} = 0.69$, and $\gamma_N^{\text{FARC}} = 0.74$. In Scenario I (DC), γ_t^* is located on average around $\bar{\gamma}^* \approx 0.564$. In Scenario II

(PAR), pressure reaches $\bar{\gamma}^* \approx 0.662$. In Scenario III (ELN), average pressure is $\bar{\gamma}^* \approx 0.663$, and in Scenario IV (FARC), it is $\bar{\gamma}^* \approx 0.715$. In all cases, γ_t^* does not identically replicate either of the pure branches from the beginning; the policy fluctuates between both benchmarks under dual control according to the material risk of the episode and the value of separating types. Nonetheless, as the posterior belief μ_t converges toward the captor's true type, the optimal instrument naturally converges to the full-information benchmark corresponding to the true perpetrator in the executed branch, here systematically approaching the Negotiation benchmark.

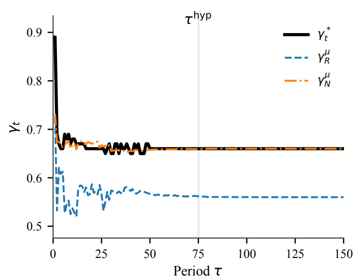
Just as with operational pressure, the optimal financial interdiction summarizes the intensity with which the State restricts payment capacity and economic coordination during the episode. Figure VII compares the trajectory of α_t^* with the belief-weighted benchmarks of the Rescue (α_R^μ) and Negotiation (α_N^μ) branches. In the baseline, the rescue benchmark values by type are $\alpha_R^{\mathrm{DC}} = 0.18$, $\alpha_R^{\mathrm{PAR}} = 0.31$, $\alpha_R^{\mathrm{ELN}} = 0.45$, and $\alpha_R^{\mathrm{FARC}} = 0.59$, while those for negotiation are $\alpha_N^{\mathrm{DC}} = 0.62$, $\alpha_N^{\mathrm{PAR}} = 0.67$, $\alpha_N^{\mathrm{ELN}} = 0.72$, and $\alpha_N^{\mathrm{FARC}} = 0.77$. In Scenario I (DC), $\bar{\alpha}^* \approx 0.896$. In Scenario II (PAR), $\bar{\alpha}^* \approx 0.730$. In Scenario III (ELN), $\bar{\alpha}^* \approx 0.737$, and in Scenario IV (FARC), $\bar{\alpha}^* \approx 0.755$. In all cases, α_t^* does not identically replicate either of the pure branches from the beginning; the policy fluctuates between both benchmarks under dual control according to accumulated information, the family's payment capacity, and the value of inducing separation among types. In scenarios of strict state capacity, the need to separate types and prevent family collusion can temporarily place interdiction above both benchmarks. Nonetheless, as the posterior belief μ_t converges toward the captor's true type, the optimal instrument naturally converges to the full-information benchmark corresponding



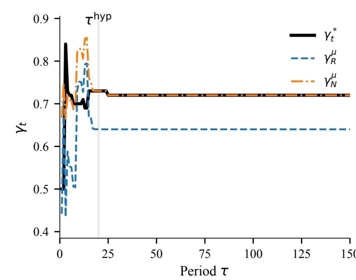
(a) Scenario I: DC



(b) Scenario II: PAR



(c) Scenario III: ELN



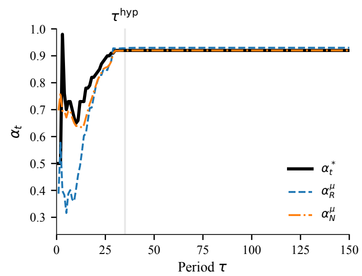
(d) Scenario IV: FARC

Figure VI: Operational pressure γ_t^* and benchmarks γ_R^μ and γ_N^μ against τ.

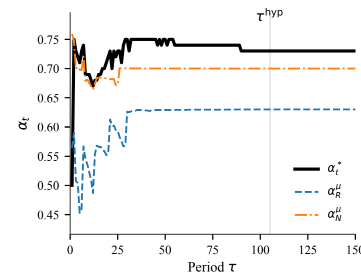
Note: Panel (a) shows the operational pressure trajectory in Scenario I (DC). Panel (b) shows the equivalent trajectory in Scenario II (PAR). Panel (c) shows Scenario III (ELN). Panel (d) shows Scenario IV (FARC). The dashed lines correspond to the Rescue (γ_R^μ) and Negotiation (γ_N^μ) belief-weighted benchmarks. The gray band marks τ^{hyp}. *Source:* Author's elaboration.

to the true perpetrator in the executed branch, here systematically approaching the Negotiation benchmark.

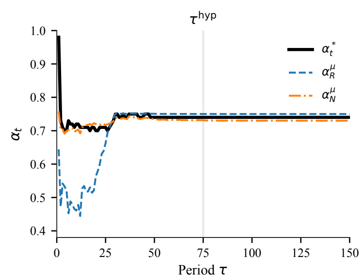
The posterior precision, defined as $\iota_t = \max_{\theta \in \Theta_K} \mu_t(\theta)$, allows observing how operational pressure changes when the state gains confidence about the identity of the true type. Figure VIII reorders the simulation on the plane (ι_t, γ_t^*); the arrows indicate the advance of τ and avoid interpreting the observations as a linear interpolation. The trajectory concentrates in zones of high pressure and rising pre-



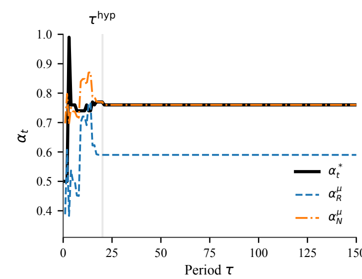
(a) Scenario I: DC



(b) Scenario II: PAR



(c) Scenario III: ELN



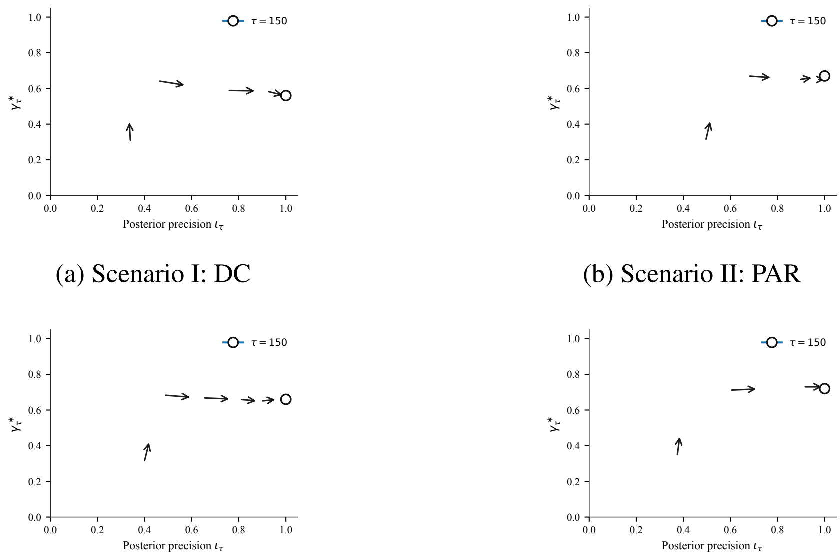
(d) Scenario IV: FARC

Figure VII: Financial interdiction α_t^* and benchmarks α_R^μ and α_N^μ against τ.

Note: Panel (a) presents the optimal financial interdiction in Scenario I (DC). Panel (b) presents Scenario II (PAR). Panel (c) presents Scenario III (ELN). Panel (d) presents Scenario IV (FARC). The dashed lines correspond to the Rescue (α_R^μ) and Negotiation (α_N^μ) belief-weighted benchmarks. The gray band marks τ^{hyp}. *Source:* Author's elaboration.

cision, reflecting learning without immediate collapse toward a single dominant signal, and showing the distinct dynamics across cases.

Financial interdiction against posterior precision allows evaluating whether learning reduces or reinforces the need to restrict payment capacity. As reported in Figure IX, α_t^* remains high while ι_t increases across all scenarios, indicating that identifying the true type does not eliminate the value of maintaining preventive interdiction. This pattern is consistent with environments where financial



(a) Scenario I: DC

(b) Scenario II: PAR

(c) Scenario III: ELN

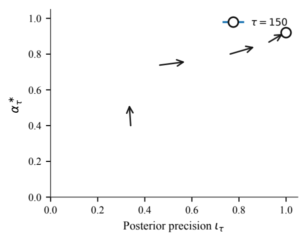
(d) Scenario IV: FARC

Figure VIII: Trajectory (ι_t, γ_t^*) with arrows indicating temporal advance.

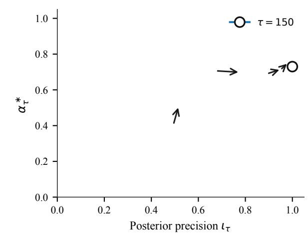
Note: Panel (a) corresponds to Scenario I (DC), panel (b) to Scenario II (PAR), panel (c) to Scenario III (ELN), and panel (d) to Scenario IV (FARC). Each trajectory shows 20 arrows indicating direction; the circle identifies the end point, $\tau = 150$. *Source:* Author's elaboration.

interdiction retains relevance as an instrument of control, payment containment, and the induction of separation among types.

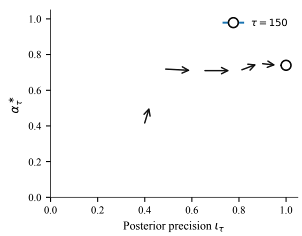
The constraints diagnostic confirms that the strategic feasibility of the mechanism is preserved across all scenarios. As Figure X shows, the captor's incentive constraint (IC^K), the captor's participation constraint (IR^K), and the family's participation constraint (IR^F) are satisfied in all 150 simulated cycles in all four scenarios. The verification is *a posteriori* with respect to the true type θ_K^*, in that the optimal instruments $(a_S^*, \alpha_t^*, \gamma_t^*)$ are evaluated directly on the utilities of the true type, without weighting by the belief μ_t. The mechanism is therefore feasible in an ex-post sense, since optimal policy, given θ_K^*, satisfies incentive compatibility



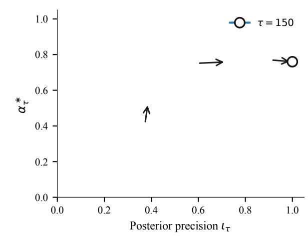
(a) Scenario I: DC



(b) Scenario II: PAR



(c) Scenario III: ELN



(d) Scenario IV: FARC

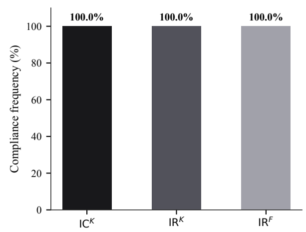
Figure IX: Trajectory (ι_t, α_t^*) with arrows indicating temporal advance.

Note: Panel (a) corresponds to Scenario I (DC), panel (b) to Scenario II (PAR), panel (c) to Scenario III (ELN), and panel (d) to Scenario IV (FARC). Each trajectory shows 20 arrows indicating direction; the circle identifies the end point, $\tau = 150$. *Source:* Author's elaboration.

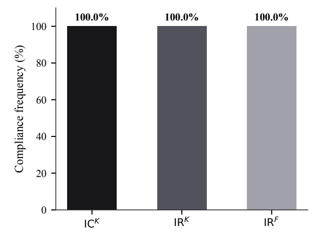
and participation without abandoning the informational component.

The expected informational gain ΔH_t quantifies the anticipated reduction in entropy induced by the state's instruments, in the sense of the optimal dynamic acquisition of information of Zhong (2022). That informational subsidy concentrates at the start of the simulation, as illustrated in Figure XI: in DC it reaches its maximum at $\tau = 3$ with $\Delta H_3 \approx 0.028$, and in PAR at $\tau = 2$ with $\Delta H_2 \approx 0.027$. As ΔH_t converges to zero, additional observations cease to reduce posterior uncertainty and the optimal decision comes to be governed by material losses and a belief already concentrated on the true type.

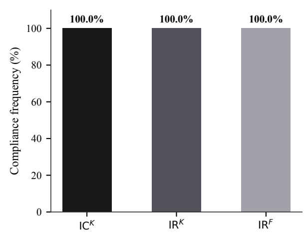
Unlike a uniform relationship, the sign of the net sensitivity $\kappa_h(\theta_K^*, t)$ varies



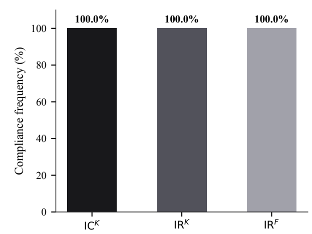
(a) Scenario I: DC



(b) Scenario II: PAR



(c) Scenario III: ELN

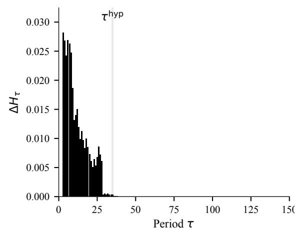


(d) Scenario IV: FARC

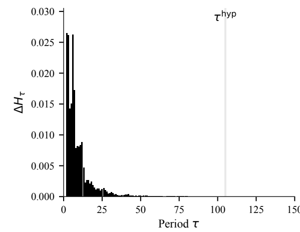
Figure X: Frequency of compliance with IC^K, IR^K, and IR^F (% of cycles).

Note: Panel (a) summarizes compliance in Scenario I (DC), panel (b) in Scenario II (PAR), panel (c) in Scenario III (ELN), and panel (d) in Scenario IV (FARC). Percentages are computed over the simulated cycles of each scenario. *Source:* Author's elaboration.

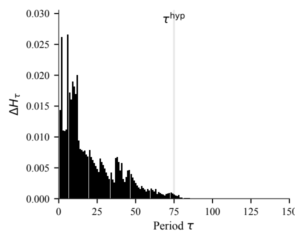
across scenarios, as illustrated in Figure XII. In Scenario II (PAR) and Scenario IV (FARC), $\kappa_h > 0$ in all or the large majority of cycles (100 % and 91.3 %, respectively), which implies $\partial\,\mathbb{E}[\tau \mid \theta_K]/\partial\gamma_t^* < 0$, meaning that higher operational pressure shortens captivity. In contrast, in Scenario I (DC) the sensitivity is predominantly negative ($\kappa_h < 0$ in 82.0 % of periods), which implies that higher pressure prolongs captivity by inhibiting the direct payment channel without sufficiently compensating through rescues. In Scenario III (ELN), the trajectory fluctuates between both regimes. The model thus illustrates a strategic trade-off, whereby pressure shortens captivity only when the risk of prolongation from negotiation deterrence is dominated by the effectiveness of rescue or capitulation.



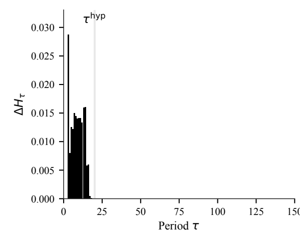
(a) Scenario I: DC



(b) Scenario II: PAR



(c) Scenario III: ELN



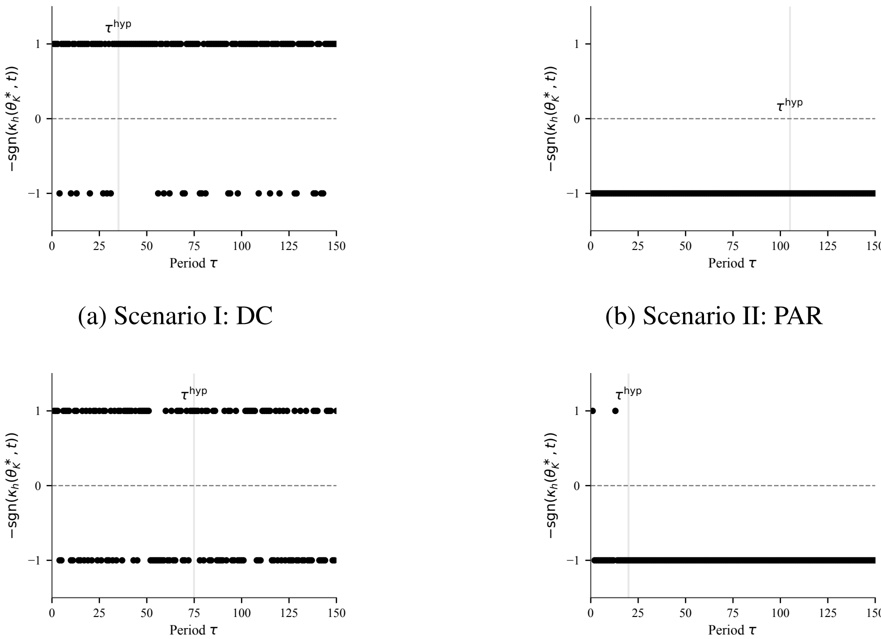
(d) Scenario IV: FARC

Figure XI: Expected informational gain ΔH_t against τ.

Note: Panel (a) corresponds to Scenario I (DC), panel (b) to Scenario II (PAR), panel (c) to Scenario III (ELN), and panel (d) to Scenario IV (FARC). The series reports expected entropy reduction in each simulated cycle. *Source:* Author's elaboration.

The hypothetical terminal outcomes differ across scenarios, which comprise *Ransom Paid* in DC ($\tau^{\text{hyp}} = 35$), *Rescue* in PAR ($\tau^{\text{hyp}} = 105$), *Release* in ELN ($\tau^{\text{hyp}} = 75$), and *Rescue* in FARC ($\tau^{\text{hyp}} = 20$). In all cases the posterior starts from its initial prior and converges to $\mu_{150}(\theta^*) \approx 1{,}000$, with $\bar{\alpha}^*$ and $\bar{\gamma}^*$ above 0,56 in all scenarios. These results are summarized in Table III.

Overall, this contrast shows that the same mechanism generates distinct learning trajectories, instrument intensities, and constraint viability rates when the true type and the institutional, family, victim, and geographic margins change. The simulations do not substitute for Theorem 8, which requires recurrent separation in the event $\mathcal{T}$; they illustrate that, with priors and hazards calibrated to Colombian



(a) Scenario I: DC

(b) Scenario II: PAR

(c) Scenario III: ELN

(d) Scenario IV: FARC

Figure XII: Net sensitivity $\kappa_h(\theta_K^*, t)$ of expected captivity duration with respect to γ_t^*.

Note: Each panel traces $\kappa_h(\theta_K^*, t)$ for the true type of the scenario. A constant value of $+1$ implies $-\operatorname{sgn}(\kappa_h) = -1$ in 100 % of cycles, that is, $\partial\,\mathbb{E}[\tau \mid \theta_K]/\partial\gamma_t^* < 0$; the horizontal dashed line marks the reference zero. *Source:* Author's elaboration.

evidence, the entropic subsidy shifts policy away from purely myopic regimes and accelerates or disciplines posterior concentration in finite horizons, in the direction predicted by the identification result.

6. Simulations under Excluded Voice Communication

To evaluate the robustness and sufficiency of the identification mechanism, this section analyzes the dynamic simulations under the counterfactual exclusion of the voice communication channel, differing from Section 5 by completely elim-

Table III: Summary of dynamic calibration by scenario

θ_K^*	$\mu_0(\theta^*)$	$\mu_{\tau_{\max}}(\theta^*)$	$\bar{\alpha}^*$	$\bar{\gamma}^*$	τ^{hyp}	$m_{\tau^{\text{hyp}}}$
DC	0.350	1.000	0.896	0.564	35	Ransom Paid
PAR	0.150	1.000	0.730	0.662	105	Rescue
ELN	0.250	1.000	0.737	0.663	75	Release
FARC	0.150	1.000	0.755	0.715	20	Rescue

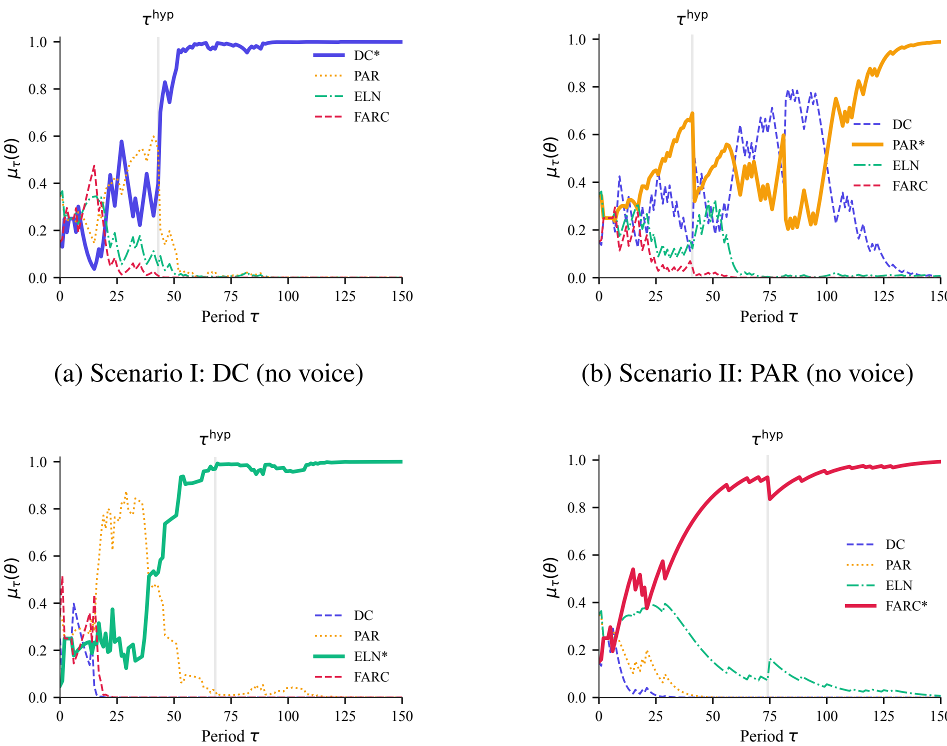
Source: Author's elaboration.

inating the acoustic information through the condition $V_t = 0$ for all t. Whereas in the baseline calibration of Section 5 the State's Bayesian update incorporates both the physical-institutional signals and the voice calls, here we isolate the endogenous physical-institutional channels by deactivating the communication likelihood feedback. Under this configuration, the only source of learning for the State consists of the physical-institutional block $\mathcal{L}_{F,t}$, which integrates the signals of the mechanism's competing risks λ_j and collusion detection d_t conditional on the actions executed after the MDG filter. The objective is to verify whether the posterior belief μ_t converges to the true captor type θ_K^* based solely on these outcomes, validating that the mechanism is self-sufficient for identification.

As shown in Figure XIII, the posterior belief of the true type $\mu_t(\theta_K^*)$ progressively approaches certainty in all four scenarios, albeit at a slower pace than in the baseline case. Within the $\tau = 150$-period horizon illustrated in the figures, the belief about the true type in Scenario I (DC) settles into near-total certainty from the earliest periods, while in Scenario II (PAR) and Scenario III (ELN) convergence is evident before the end of the illustrated period. In Scenario IV (FARC), by contrast, learning advances gradually, indicating that full convergence requires longer simulation horizons, on the order of 400 periods, extending beyond the visual limit of the figures. These results illustrate finite-horizon behavior in the direction predicted by Theorem 8, without constituting a direct asymptotic validation, given that the physical-institutional competing risks (λ_j), which depend on the type-contingent coefficients and on the State's controls (α_t, γ_t), exhibit sufficient variation to asymptotically separate the types. Deactivating the voice channel does not block identification, but it significantly slows the speed of convergence relative to Section 5. In the baseline case, the acoustic features allowed the posterior to concentrate within 20 to 105 periods; without them, the State requires a substantially longer history of physical-institutional signals to achieve equivalent precision. This highlights the role of voice as an informational accelerator, rather than as a necessary condition for identification.

Upon excluding voice feedback, the trajectory of the optimal operational pressure γ_t^* and its comparison with the γ_R^μ and γ_N^μ benchmarks, detailed in Figure XIV, evidence the impact of slower learning on instrument persistence. Because the acquisition of information is slower without acoustic feedback, the State is forced to prolong elevated levels of operational pressure for a larger number of periods in order to induce separation and force revelation through the rescue hazard. For instance, in the DC case, average pressure stands at $\bar{\gamma}^* \approx 0.577$ across the 150 simulated periods, compared with the $\bar{\gamma}^* \approx 0.564$ recorded in Section 5. As in the baseline specification, dual control causes the pressure trajectory to fluctuate between both benchmarks before converging toward the benchmark of the executed branch (Negotiation) once uncertainty has dissipated, although the absence of voice signals significantly prolongs the active-learning phase.

In a similar vein, the trajectory of the optimal financial interdiction α_t^* relative



(a) Scenario I: DC (no voice)

(b) Scenario II: PAR (no voice)

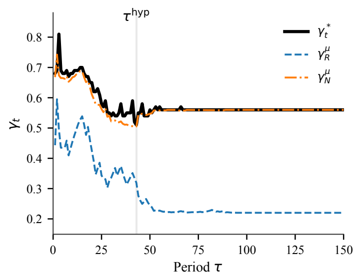
(c) Scenario III: ELN (no voice)

(d) Scenario IV: FARC (no voice)

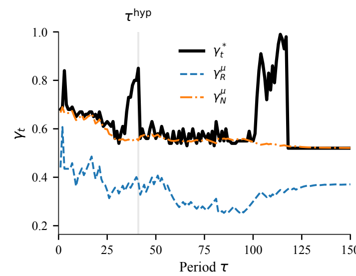
Figure XIII: Bayesian beliefs of each type $\mu_\tau(\theta)$ across scenarios under excluded voice communication.

Note: Panels (a)-(d) replicate the baseline scenarios but with the voice communication channel deactivated. The gray band marks the counterfactual stopping boundary τ^{hyp}. *Source:* Author's elaboration.

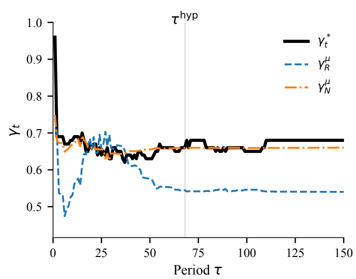
to the α_R^μ and α_N^μ benchmarks, illustrated in Figure XV, reflects an equivalent persistence under the absence of communication. As with operational pressure, the State is forced to sustain interdiction at elevated levels for a larger absolute number of periods across all no-voice scenarios, thereby compensating for the lack of acoustic signals and avoiding premature negotiation. For example, in Scenario I (DC) this greater absolute persistence does not translate into a higher average,



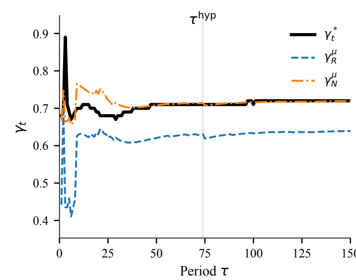
(a) Scenario I: DC (no voice)



(b) Scenario II: PAR (no voice)



(c) Scenario III: ELN (no voice)



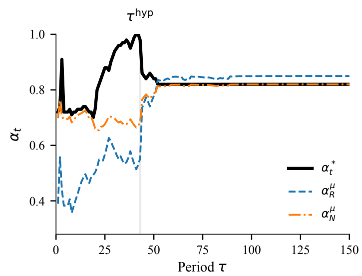
(d) Scenario IV: FARC (no voice)

Figure XIV: Operational pressure γ_t^* and benchmarks γ_R^μ and γ_N^μ against τ under excluded voice communication.

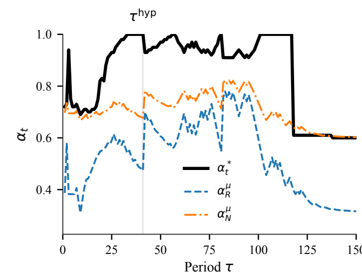
Note: Panels (a)-(d) replicate the baseline scenarios but with the voice communication channel deactivated. The dashed lines correspond to the Rescue (γ_R^μ) and Negotiation (γ_N^μ) belief-weighted benchmarks. The gray band marks the counterfactual stopping boundary τ^{hyp}. *Source:* Author's elaboration.

since the gradual withdrawal of the instrument after the late convergence extends the relevant horizon out to the full 150 simulated periods, over which the average is $\bar{\alpha}^* \approx 0.827$; the baseline case with voice, by contrast, converges in only 35 periods and records a higher average of 0.896, precisely because that shorter window does not reach the decline phase that follows convergence.

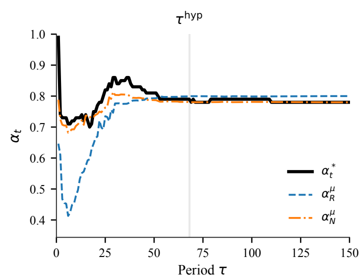
The trajectories of posterior precision and operational pressure in the joint space (ι_t, γ_t^*), shown in Figure XVI, reveal a pattern distinct from that of Sec-



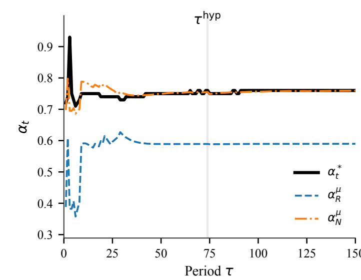
(a) Scenario I: DC (no voice)



(b) Scenario II: PAR (no voice)



(c) Scenario III: ELN (no voice)



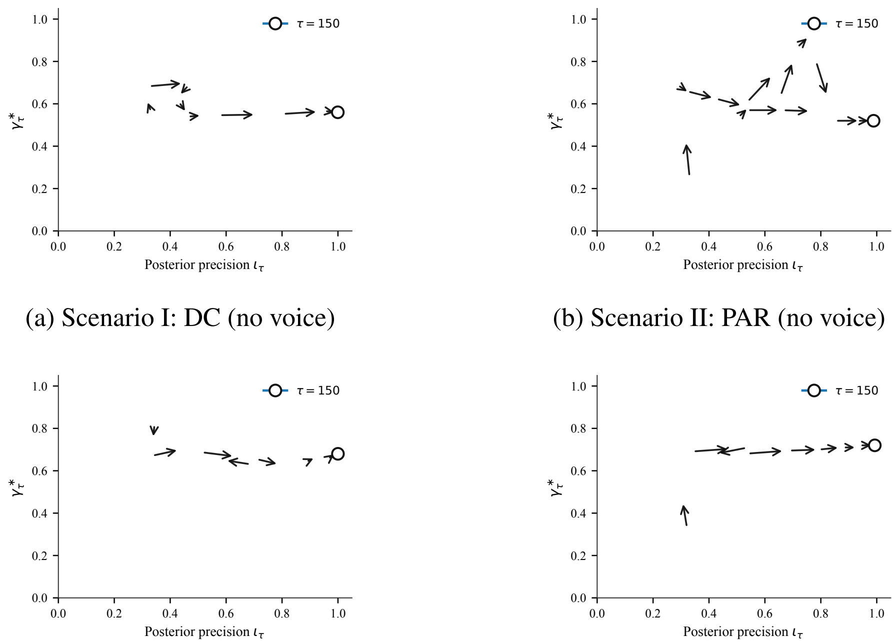
(d) Scenario IV: FARC (no voice)

Figure XV: Financial interdiction α_t^* and benchmarks α_R^μ and α_N^μ against τ under excluded voice communication.

Note: Panels (a)-(d) replicate the baseline scenarios but with the voice communication channel deactivated. The dashed lines correspond to the Rescue (α_R^μ) and Negotiation (α_N^μ) belief-weighted benchmarks. The gray band marks the counterfactual stopping boundary τ^{hyp}.

Source: Author's elaboration.

tion 5. Whereas there voice feedback and the optimal perturbation of the State's instruments generated marked informational jumps in the early periods, here posterior precision rises progressively across all scenarios, driven solely by the optimal perturbation of the mechanism through the pressure and control instruments (γ_t^* and α_t^*). This dynamic accumulation of precision reflects that learning rests on the separation of the MDG hazard rates under the State's strategic inducement, without the aid of discrete acoustic shocks. Consequently, operational pressure is gradually reduced as precision converges to its upper bound, illustrating the tight coupling between the accumulation of physical information and the withdrawal of the instrument.



(a) Scenario I: DC (no voice)

(b) Scenario II: PAR (no voice)

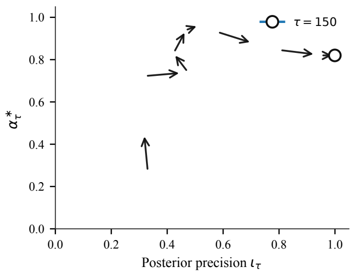
(c) Scenario III: ELN (no voice)

(d) Scenario IV: FARC (no voice)

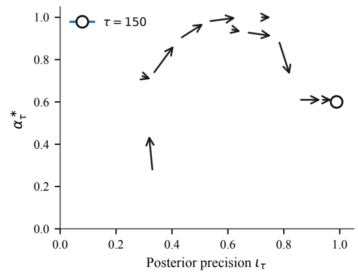
Figure XVI: Posterior precision and operational pressure trajectory (ι_t, γ_t^*) under excluded voice communication.

Source: Author's elaboration.

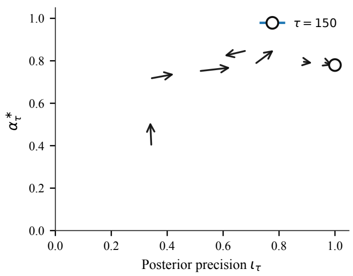
The joint trajectories of posterior precision and financial interdiction (ι_t, α_t^*), shown in Figure XVII, display a pattern analogous to that of operational pressure, in which the withdrawal of financial interdiction toward the optimal full-information branch is much slower and smoother. The absence of the acoustic informational subsidy eliminates rapid policy transitions, forcing the State to sustain financial interdiction at high levels in order to preserve incentive compatibility and thereby avoid capitulation, while the belief updates slowly, period by period, from the physical outcomes.



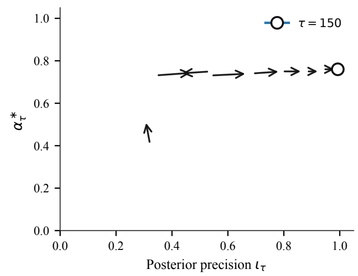
(a) Scenario I: DC (no voice)



(b) Scenario II: PAR (no voice)



(c) Scenario III: ELN (no voice)

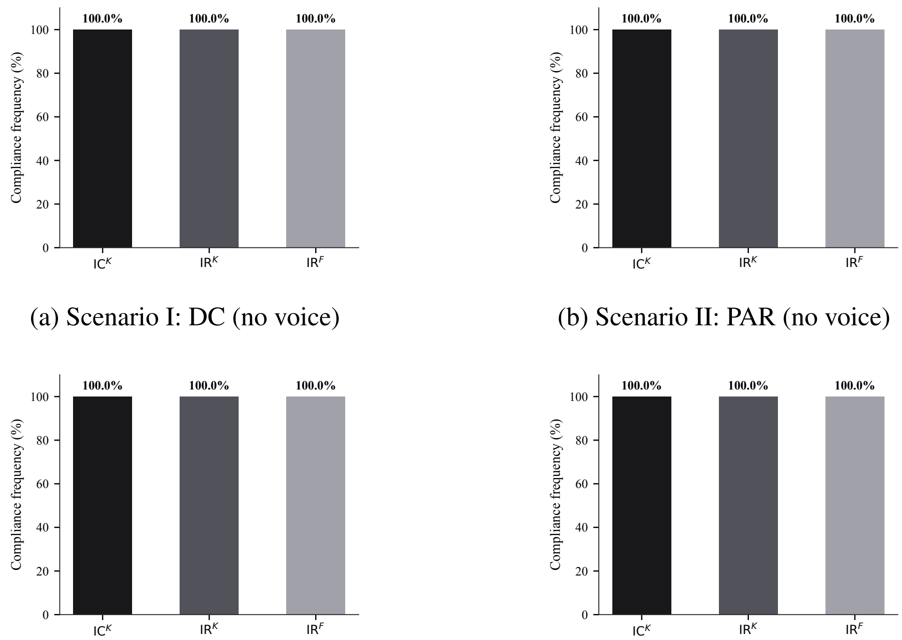


(d) Scenario IV: FARC (no voice)

Figure XVII: Posterior precision and financial interdiction trajectory (ι_t, α^*_t) under excluded voice communication.

Source: Author's elaboration.

The compliance diagnostics in Figure XVIII confirm that the mechanism's strategic feasibility is preserved at 100 % across all simulated cycles, even under the slow-update conditions caused by the exclusion of voice. The family's and the captor's individual-rationality constraints, denoted IR^F and IR^K, together with the captor's incentive-compatibility constraint, IC^K, are satisfied without exception. This demonstrates that the State's dual-control formulation is robust to changes in the speed of learning, adjusting the policy trajectories to ensure that the equilibrium's participation and incentive constraints remain feasible in every period.



(a) Scenario I: DC (no voice)

(b) Scenario II: PAR (no voice)

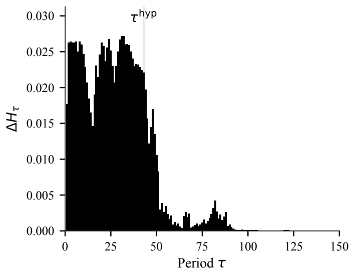
(c) Scenario III: ELN (no voice)

(d) Scenario IV: FARC (no voice)

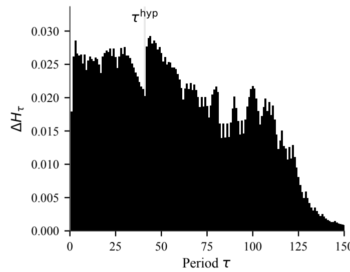
Figure XVIII: Frequency of compliance with IC^K, IR^K, and IR^F under excluded voice communication.

Source: Author's elaboration.

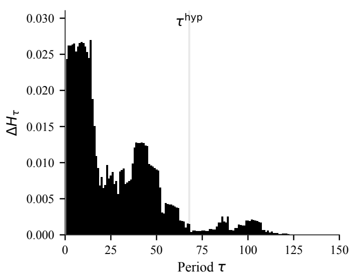
The expected informational gain ΔH_t under excluded voice communication, illustrated in Figure XIX, extends over a considerably longer horizon than in Section 5. Without the acoustic signals that accelerated convergence, the State depends exclusively on the physical-institutional signals of competing risks and detection, and the informational subsidy remains positive for a larger number of periods before converging to zero. This pattern is consistent with the longer duration observed in the concentration of posterior beliefs under voice exclusion, and confirms that the active-exploration motive continues to operate even when the only available channel is the physical-institutional one.



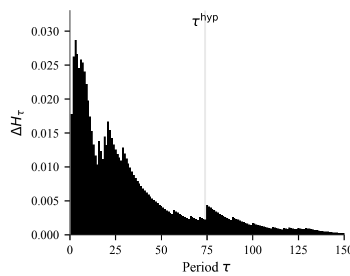
(a) Scenario I: DC (no voice)



(b) Scenario II: PAR (no voice)



(c) Scenario III: ELN (no voice)

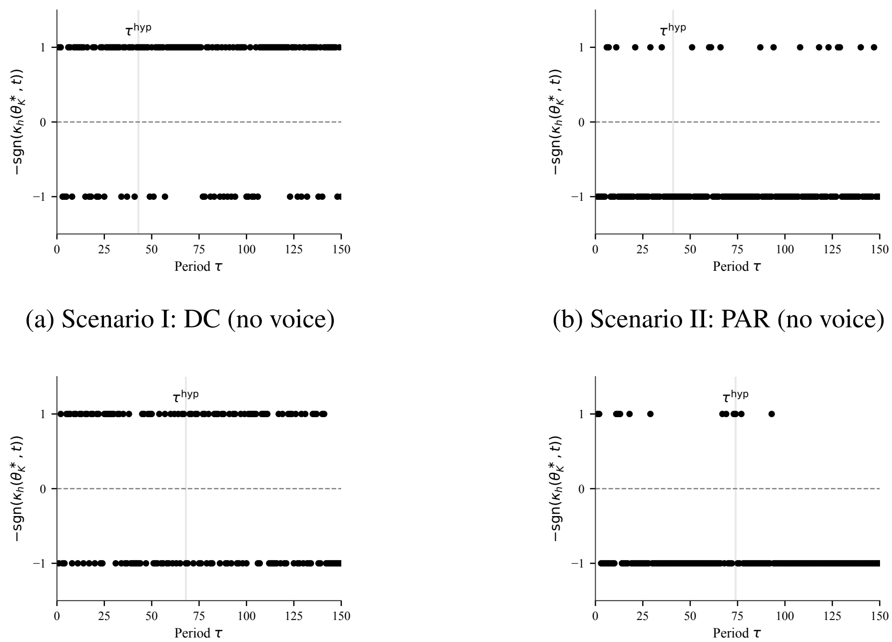


(d) Scenario IV: FARC (no voice)

Figure XIX: Expected informational gain ΔH_t against τ under excluded voice communication.

Source: Author's elaboration.

The net sensitivity of expected captivity duration $\kappa_h(\theta_K^*, t)$ to operational pressure, plotted in Figure XX, reproduces the same heterogeneous pattern across scenarios identified in Section 5. In Scenario II (PAR) and Scenario IV (FARC), $\kappa_h > 0$ in $87.3\,\%$ and $91.3\,\%$ of cycles, respectively, implying $\partial\,\mathbb{E}[\tau \mid \theta_K]/\partial\gamma_t^* < 0$. In contrast, in Scenario I (DC) the sensitivity is again predominantly negative, with $\kappa_h < 0$ in $74.7\,\%$ of periods, so that greater operational pressure tends to prolong captivity. In Scenario III (ELN), the trajectory is split almost evenly between both regimes, with $\kappa_h > 0$ in $50.0\,\%$ of periods. This result confirms that, even in the absence of the voice channel, the strategic trade-off between operational pressure and expected captivity duration identified in the baseline calibration remains without qualitative alterations.



(a) Scenario I: DC (no voice)

(b) Scenario II: PAR (no voice)

(c) Scenario III: ELN (no voice)

(d) Scenario IV: FARC (no voice)

Figure XX: Net sensitivity $\kappa_h(\theta_K^*, t)$ of expected captivity duration with respect to γ_t^* under excluded voice communication.

Source: Author's elaboration.

The quantitative results of these counterfactual simulations, evaluated at the same baseline reference stopping period τ^{hyp}, are summarized in Table IV. At the reference stopping point, the posterior belief of the true type concentrates slowly in DC, reaching only $\mu_{35} \approx 0.222$; in PAR it reaches $\mu_{105} \approx 0.718$; and in FARC the slower learning results in a belief of only $\mu_{20} \approx 0.463$; while in ELN the belief rises to $\mu_{75} \approx 0.989$. This illustrates the delay in the speed of posterior concentration induced by the loss of acoustic information, forcing a much longer interaction horizon before closure is achieved in most scenarios.

Table IV: Summary of dynamic calibration under excluded voice communication

θ_K^*	$\mu_0(\theta^*)$	$\mu_{\tau^{\text{hyp}}}(\theta^*)$	$\bar{\alpha}^*$	$\bar{\gamma}^*$	τ^{hyp}	$m_{\tau^{\text{hyp}}}$
DC	0.150	0.222	0.806	0.639	35	Ransom Paid
PAR	0.350	0.718	0.913	0.617	105	Rescue
ELN	0.050	0.989	0.791	0.662	75	Release
FARC	0.150	0.463	0.751	0.706	20	Rescue

Source: Author's elaboration.

7. Discussion and conclusions

This paper designs an indirect dynamic mechanism for environments where equilibrium behavior pools and types are not reported. Relative to the classical reduction via the Revelation Principle, the contribution is to construct the operational indirect object, composed of public instruments, endogenous implementation noise, and Bayesian learning over action-generated signals.

At the theoretical level, the MDG kernel supplies full-support materialization of latent intentions without an exogenous tremble, keeping Bayes' rule well defined off path. The Rational Type Identification Theorem shows that recurrent total-variation separation of type-contingent observation laws implies mutual singularity of trajectory measures and almost-sure posterior concentration on the true type on the prolonged-interaction event $\mathcal{T}$. The Conditional Implementability Theorem complements identification by guaranteeing a PBE that satisfies incentive compatibility and individual rationality whenever the feasible set remains non-empty.

At the applied level, Colombian kidnapping for ransom maps the abstract type space into estimated criminal technologies, where competing risk and multinomial choice evidence discipline the hazards and the priors. The dynamic simulations of the baseline scenarios show that the calibrated dual-control rule accelerates posterior concentration on the true type, sustains elevated but disciplined instrument intensities, and preserves the feasibility of the IR and IC constraints, in the direction predicted by the theory. Likewise, the counterfactual analysis under exclusion of the voice communication channel validates the robustness of the mechanism, showing that Bayesian identification is self-sufficient based solely on the physical-institutional signals of competing risks and detection, although the speed of posterior convergence slows significantly and demands a longer interaction history to achieve an equivalent level of precision.

A research agenda consists of extending the Rational Type Identification Theorem to other economic environments where observational pooling prevents the passive discrimination of heterogeneous agents. In the labor market, where workers with different latent productivities (θ) choose the same equilibrium level of education or effort, the MDG process can be formalized as a test design or stochastic task assignment with endogenous noise that perturbs intentions and forces the emission of full-support signals. In financial intermediation, facing borrowers with varying default probabilities who mimic low-risk schemes, future work can adapt the mechanism to channel credit applications through noisy audits and dynamic incentives, separating risk profiles through the lender's Bayesian convergence. Similarly, in goods markets with asymmetric quality, the MDG architecture allows decomposing firms' signaling strategies, such as prices, warranties, or certifications, into full-support variations. On all three fronts, evaluating the total-variation separation of the resulting observation laws will make it possible to verify how the market filtration $\mathcal{F}_t^{\mathrm{pub}}$ refines the Principal's beliefs, guaranteeing almost-sure convergence toward the agent's true identity via Kakutani (1948).

Declaration on the use of Generative Artificial Intelligence

In the preparation of this manuscript, the author used generative artificial intelligence tools, specifically *Cursor* and *Antigravity*, to assist in tasks such as literature review and text editing; programming in Lean 4 for the formalization and interactive proof of theoretical theorems and lemmas; programming of quantitative and empirical code in Matlab, Stata, and Python; as well as the design and online deployment of the calibrated dynamic model interface via GitHub and Streamlit. Consistent with academic ethical guidelines, the author is solely responsible for the integrity of the document, having thoroughly verified and validated all information, proofs, estimations, and codes provided or generated by these technologies.

References

Abbring, Jaap H. and Gerard J. Van den Berg (2003), "The nonparametric identification of treatment effects in duration models." *Econometrica*, 71 (5), 1491–1517.

Aghion, Philippe, Patrick Bolton, Christopher Harris, and Bruno Jullien (1991), "Optimal learning by experimentation." *The Review of Economic Studies*, 58 (4), 621–654.

Becker, Gary S. (1968), "Crime and punishment: An economic approach." *Journal of Political Economy*, 76 (2), 169–217.

Bergemann, Dirk and Stephen Morris (2012), "An introduction to robust mechanism design." *Foundations and Trends in Microeconomics*, 8 (3), 169–230.

Bergemann, Dirk and Juuso Välimäki (2019), "Dynamic mechanism design: An introduction." *Journal of Economic Literature*, 57 (2), 235–274.

Blackwell, David and Lester E. Dubins (1962), "Merging of opinions with increasing information." *The Annals of Mathematical Statistics*, 33 (3), 882–886.

Bolton, Patrick and Mathias Dewatripont (2005), *Contract Theory*, MIT Press, Cambridge, MA.

Bradburn, MJ, TG Clark, SB Love, and DG Altman (2003), "Survival analysis part ii: Multivariate data analysis – an introduction to concepts and methods." *British Journal of Cancer*, 89 (3), 431–436.

CNMH (2013), *Una sociedad secuestrada*, first edition, Imprenta Nacional, Bogotá, Colombia. Centro de Memoria Histórica.

CNMH (2026), "Portal de datos: El conflicto en cifras." Conflicto en Cifras.

Courty, Pascal and Hao Li (2000), "Sequential screening." *Review of Economic Studies*, 67 (4), 697–717.

Cox, D. R. (1972), "Regression models and life-tables." *Journal of the Royal Statistical Society. Series B (Methodological)*, 34 (2), 187–220.

CRC, Law 282 (1996), "Law 282 de 1996." Diario Oficial n.º 42.802. Eva Función Pública.

Doob, Joseph L. (1953), *Stochastic Processes*, Wiley, New York. Disponible en Internet Archive.

Easley, David and Nicholas M. Kiefer (1988), "Controlling a stochastic process with unknown parameters." *Econometrica*, 56 (5), 1045–1064.

Eső, Péter and Balázs Szentes (2007), "Optimal information disclosure in auctions and the handicap auction." *Review of Economic Studies*, 74 (3), 705–731.

Fudenberg, Drew and David K. Levine (1993), "Self-confirming equilibrium." *Econometrica*, 61 (3), 523–545.

Fudenberg, Drew and Jean Tirole (1991a), *Game Theory*, MIT Press, Cambridge, MA.

Fudenberg, Drew and Jean Tirole (1991b), "Perfect bayesian equilibrium and sequential equilibrium." *Journal of Economic Theory*, 53 (2), 236–260.

Gibbard, Allan (1973), "Manipulation of voting schemes: A general result." *Econometrica*, 41 (4), 587–601.

Holmström, Bengt and Roger B. Myerson (1983), "Efficient and durable decision rules with incomplete information." *Econometrica*, 51 (6), 1799–1819.

Jaynes, E. T. (1957), "Information theory and statistical mechanics." *Physical Review*, 106 (4), 620–630.

Kadan, Ohad, Philip J. Reny, and Jeroen M. Swinkels (2017), "Existence of optimal mechanisms in principal-agent problems." *Econometrica*, 85 (3), 769–823.

Kakutani, Shizuo (1948), “On equivalence of infinite product measures.” *Annals of Mathematics*, 49 (2), 214–224.

Kalai, Ehud and Ehud Lehrer (1993), “Rational learning leads to nash equilibrium.” *Econometrica*, 61 (5), 1019–1045.

Kreps, David M. and Robert Wilson (1982), “Sequential equilibria.” *Econometrica*, 50 (4), 863–894.

Lancaster, Tony (1990), *The Econometric Analysis of Transition Data*, Cambridge University Press, Cambridge.

Lucas, Robert E. Jr. (1976), “Econometric policy evaluation: A critique.” *Carnegie–Rochester Conference Series on Public Policy*, 1, 19–46.

Magnac, Thierry and David Thesmar (2002), “Identifying dynamic discrete decision processes.” *Econometrica*, 70 (2), 801–816.

McCall, John J. (1970), “Economics of information and job search.” *The Quarterly Journal of Economics*, 84 (1), 113–126.

Myerson, Roger B. (1979), “Incentive compatibility and the bargaining problem.” *Econometrica*, 47 (1), 61–73.

Myerson, Roger B. (1981), “Optimal auction design.” *Mathematics of Operations Research*, 6 (1), 58–73.

Myerson, Roger B. and Mark A. Satterthwaite (1983), “Efficient mechanisms for bilateral trading.” *Journal of Economic Theory*, 29 (2), 265–281.

Pavan, Alessandro, Ilya Segal, and Juuso Toikka (2014), "Dynamic mechanism design: A myersonian approach." *Econometrica*, 82 (2), 601–653.

Pinto, María Eugenia, Ivette Ma. Altamar, Yilberto Lahuerta, Luis Fernando Cepeda, and Adriana Victoria Mera (2004), "El secuestro en colombia: Caracterización y costos económicos." Technical Report 257, Departamento Nacional de Planeación, Bogotá.

Pires, Stephen F., Rob T. Guerette, and Christopher H. Stubbert (2014), "The crime triangle of kidnapping for ransom incidents in colombia, south america: A "litmus" test for situational crime prevention." *The British Journal of Criminology*, 54 (5), 784–808.

Rochet, Jean-Charles (1987), "A necessary and sufficient condition for rationalizability in a quasi-linear context." *Journal of Mathematical Economics*, 16 (2), 191–200.

Rochet, Jean-Charles and Philippe Choné (1998), "Ironing, sweeping, and multidimensional screening." *Econometrica*, 66 (4), 783–826.

Rust, John (1987), "Optimal replacement of GMC bus engines: An empirical model of Harold Zurcher." *Econometrica*, 55 (5), 999–1033.

Schervish, Mark J. (1995), *Theory of Statistics*, Springer Series in Statistics, Springer, New York.

Selten, Reinhard (1975), "Reexamination of the perfectness concept for equilibrium points in extensive games." *International Journal of Game Theory*, 4 (1), 25–55.

Shannon, Claude E. (1948), “A mathematical theory of communication.” *Bell System Technical Journal*, 27 (3), 379–423.

Shiryaev, Albert N. (2019), *Probability*, 2 edition, Graduate Texts in Mathematics, Vol. 95, Springer, New York.

Sueyoshi, Glenn T. (1992), “Semiparametric proportional hazards estimation of competing risks models with time-varying covariates.” *Journal of Econometrics*, 51 (1-2), 25–58.

Wooldridge, Jeffrey M. (2010), *Econometric Analysis of Cross Section and Panel Data*, second edition, The MIT Press, Cambridge, Massachusetts and London, England.

Zhang, Luyao and Dan Levin (2017), “Bounded rationality and robust mechanism design: An axiomatic approach.” *American Economic Review*, 107 (5), 235–239.

Zhong, Weijie (2022), “Optimal dynamic information acquisition.” *Econometrica*, 90 (4), 1537–1582.

Zárate, Mauricio Andrés (2007), *El secuestro en Colombia: una aproximación desde la teoría económica del conflicto – secuestro*. Master’s thesis, Universidad de los Andes, Facultad de Economía, Bogotá.